Supplementary Information for

Antibiotic contamination stimulates microbial methylmercury synthesis in rice paddies

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Supplementary Text

1. Impacts of molybdate doses on Hg methylation induced by FQ stress

Previous studies propose that excessive doses of molybdate might lead to an overestimation or misrepresentation of the contribution of methylators, potentially resulting in misguided strategies for managing and mitigating MeHg risks[1]. To verify whether the addition of molybdate, i.e., 20 mM, was overdosed and led to unreliable results, we designed an additional metabolism assay where 5 or 10 mM molybdate was added to four soils used in Batch assays III and IV, i.e., Tianmen, Huanggang, Meishan, and Ganzhou. The concentrations of molybdate were comparable to the total sulfur concentrations in the soil (Supplementary Table 14). The incubation process was conducted based on the description in the main text. After 21 days of incubation, soil MeHg content was analyzed to quantify the inhibitory effects of different molybdate doses. Results showed that the reductions in net MeHg production among different molybdate-dose treatments were comparable (Supplementary Fig. 6), suggesting that the molybdate dose of 20 mM was acceptable and that the results presented in the main text (Fig. 2a–d) are reliable.

Furthermore, results from other assays also support the dominant role of SRB in elevated Hg methylation. Firstly, the increases in soil MeHg were most notable in soil with high relative abundance of *hgcA*-containing SRB (MY soil; Fig. 1), while changes were minor in soil with high relative abundance of *hgcA*-containing iron-reducing bacteria (DL soil; Fig. 1) and methanogens (KL soil; Fig. 1). Secondly, four SRB genera have been identified based on the results of 16S rRNA sequencing, and three of them collectively accounted for 63% of the variance in MeHg net production. Thirdly, results from the metagenomic sequencing further revealed that it was the complete SRB with the acetyl-CoA pathway that mainly induced the increases in MeHg production under FQ stress. Collectively, despite the relatively high molybdate dose used in this study, the main conclusion that SRB are the key group for elevated Hg methylation is reliable.

2. Microbial composition of *hgcA*-containing communities in soils

We analyzed the microbial composition of the *hgcA*-containing *Deltaproteobacteria* and Archaea as described in the main text (*hgcA* gene amplicon sequencing and quantitative PCR; Supplementary Figs. 9–11 and Supplementary Tables 6–8). In general, ENR treatment had little impact on the alpha-diversity of *Deltaproteobacteria* (Shannon, Chao_1, and OTU reads; Supplementary Fig. 9). Specifically, a total of five genera were identified, including *Geobacter*, *Desulfuromonas*, *Desulfobulbus*, *Syntrophus*, and *Methanocella* from *Deltaproteobacteria* (Supplementary Fig. 10a), among which three of five were sulfate-reducing bacteria (SRB). The relative abundance of these three SRB genera did not change significantly (Supplementary Fig. 10b), suggesting that the *hgcA*-containing SRB community might also be impacted by FQ addition, i.e., these SRB genera may not resist FQ. Notably, soil MeHg contents were significantly and positively correlated with their abundance (Supplementary Fig. 10c) but not with the relative abundance of *hgcA*-containing *Geobacter* (Supplementary Fig. 11). These correlations suggest that Hg-methylating SRB genera might play a dominant role in MeHg production irrespective of the presence of ENR, while *hgcA*-containing iron-reducing bacteria (FeRB) might not.

For Archaea, although the copies of *hgcA* significantly increased upon the addition of ENR and OFL (Fig. 2j), the overall changes induced by the four FQs were not significant (data not shown). Meanwhile, the alpha diversity of *hgcA*-containing Archaea did not change significantly (Supplementary Table 6). Further analysis at the phylum level revealed that three phyla had been identified, i.e., *Euryarchaeota*, *Crenarchaeota*, and *Thaumarchaeota*, while their relative abundances were impacted by ENR stress to a minor extent (Supplementary Table 7). At the genus level, a total of nine genera were observed in the soil, and the relative abundance of the Archaeal community was not impacted by ENR addition (Supplementary Table 8). These findings collectively suggest that *hgcA*-containing Archaea play a negligible role in the elevated Hg methylation under FQ stress.

3. Results of PICRUSt2 based on 16S rRNA sequencing

Microbial communities were analyzed for the eight soil samples (paired with the control and ENR-treated sample) from four sampling sites (Meishan, Ganzhou, Huanggang, and Tianmen) using a 16S rRNA gene amplicon sequencing method. PICRUSt2 predictions showed that ENR addition to soils significantly inhibited a total of 11 functional metabolic pathways via carbon metabolism, whereas only the abundance of chitin derivative degradation significantly increased under the ENR stress (Supplementary Fig. 14a).

In particular, the reductive acetyl-coenzyme A (acetyl-CoA) pathway was significantly inhibited by the ENR treatment. The abundance of the acetyl-CoA pathway as a C1 metabolism was negatively correlated with the potential Hg methylation ($R^2 = 0.655$, $P = 0.015$, n = 8; Supplementary Fig. 14b). Among the 10 enzyme types involved in the acetyl-CoA pathway, only the formate dehydrogenase (with $NADP^+$ involved) alpha subunit (FDHA) was significantly inhibited by ENR amendments (paired t-test = 4.29, $P = 0.023$; Supplementary Fig. 14c). From the classical Wood-Ljungdahl pathway, ENR addition to soils inhibited formate oxidation, which might contribute to more methyl groups by formate metabolism (Supplementary Fig. 14d). Further analyses indicated that the abundance of the *fdhA* gene was negatively correlated with the potential Hg methylation ($R^2 = 0.56$, $P = 0.03$, n = 8; Supplementary Fig. 14e). In summary, the prediction results suggested that FQ addition to the soils might inhibit both the formate oxidation and acetyl-CoA pathway, which likely accumulates more methyl groups to synthesize MeHg.

PICRUSt2 analyses showed that two metabolic pathways were involved in sulfate reduction. The abundance of sulfate reduction I (assimilatory) was 34558 in the CK and 31263 in the ENR-spiked soils (Supplementary Fig. 15), and the abundance of the super-pathway for sulfate assimilation and cysteine biosynthesis was 40,305 in the CK and 36,878 in the ENR. In general, ENR addition did not impact either the sulfate reduction I (paired t-test = 1.58, $P = 0.21$, n = 8) or the super-pathway of sulfate assimilation and cysteine biosynthesis (paired t-test = 1.44, $P = 0.21$, n = 8). Results from 16S rRNA and PICRUSt2-based functional prediction consistently suggest an

association between SRB and FQ-enhanced MeHg production. However, because PICRUSt2 infers metabolic potential based only on phylogenetic information rather than direct functional measurements, these results should be interpreted cautiously.

4. Bin-level of *hgcAB*-harboring microbial community response to FQ stress

In the SRB-enriched cultures, metagenomic sequencing was conducted to further investigate the relationship between ENR-enhanced MeHg production and changes in the *hgcAB*-harboring microorganisms. A total of 9 metagenome-assembled genomes (MAGs) bins containing *hgcAB* were identified, all of which belonged to the phylum *Firmicutes*. Among these, a *Desulfosporosinus* MAG harboring *hgcAB* was consistently detected across all treatments (Supplementary Fig. 22a). This observation is consistent with soil-level 16S rRNA gene analysis, which identified *Desulfosporosinus* as a primary contributor to ENR-enhanced MeHg production in the paddy soil microbial community (Fig. 3h).

Bin-level abundance of *hgcAB*-containing MAGs, expressed as transcripts per million (TPM), exhibited variable response to ENR exposure, with relative change ranging from -35.6% to +75% compared to the corresponding controls (Supplementary Fig. 22b). In contrast, MeHg concentrations increased from 14.1% to 132% after 24h ENR exposure (Fig. 5g). These results indicated that Hg methylation potential was more closely associated with the metabolic activity of *hgcAB*-containing *Desulfosporosinus* than with their abundance level. Although *Desulfosporosinus* exhibited relatively high abundance in the enriched communities derived from Meishan soil, under ENR stress, Hg methylation potentials in this system remained at low levels. Furthermore, the abundance of *hgcAB-Desulfosporosinus* decreased by -35.6% under ENR exposure, yet MeHg production was not proportionally reduced, suggesting that elevated *hgcAB* abundance alone does not necessarily lead to enhanced MeHg production. Instead, methylation potential appears to be governed primarily by the metabolic activity of Hg-methylating microorganisms, even when their abundance remains low.

5. Hg methylation by enriched consortia

The Mercury methylation assay was conducted twice using the enriched consortia. Although the enriched microorganisms were cultivated using the same procedures and under identical environmental conditions (e.g., pH and temperature), the quantified Hg methylation potential varied greatly. To be specific, Hg methylation potential increased by 18.4%–108% (Supplementary Table 10) in the first trial and by 13.4- to 58.9-fold in the second trial (Supplementary Fig. 18b).

The primary reason for the wide variation in increases is that during the second performance of the Hg methylation assay, the Hg methylation in CK was very low in the second test (0.8%–2.4% vs. 4.0%–13.3% in the first trial). The highest Hg methylation potential reached 48.0% in the repeated methylation assays, whereas the lowest value was 0.8%. These values fall within the range previously reported for environmental Hg methylators under lab conditions (0.02%–54.5%)[2], indicating that the methylation capacities observed in our study are environmentally relevant and biologically plausible. Although the experimental conditions, including incubation time, Hg concentration, and temperature, were identical for these two tests, the microbial community may change during different batches of enrichment SRB from soil.

Despite the large variation in Hg methylation potential, the main results of this work are reliable, as each key result is supported by multiple lines of evidence. First, the elevated MeHg production was observed in both soil incubation experiments (Batch assays I, II, and III) and enriched consortia (Batch assay IV) under FQ exposure. Second, the dominant role of SRB in the elevated Hg methylation was supported by the metabolism inhibition assays (Batch assay III), 16S rRNA gene, and *hgcA* gene profiles. Third and most importantly, the stimulated activity of Hg methylating SRB with the acetyl-CoA pathway is supported by metagenomic analysis of SRB-enriched cultures.

Supplementary Methods

1. HCl-extractable Fe

Iron (Fe) reduction was indicated by the amount of ferrous Fe extracted using a 0.5 M HCl solution[3]. Briefly, ~0.1 g of soil and 2 mL of HCl were added to a 15 mL centrifuge tube. The slurry was shaken at room temperature at 230 rpm for 1 h. After extraction, the slurry was diluted with deoxygenated deionized water, centrifuged at 4000 rpm for 10 min, and then filtered through a 0.45 μm-filter. The ferrous iron concentration was measured according to the phenanthroline method[4]. Briefly, 1 mL of the solution sample was added to a tube containing 19 mL deoxygenated deionized water and 100 μL 10% acetate solution (to ensure a pH range of 4 to 5). After being mixed thoroughly, 0.5 mL of 0.1% phenanthroline solution was added and then subjected to absorbance analysis using a portable spectrometer (A = 550 nM) (HACH, DR1900).

2. Water-extractable THg

In this study, Hg mobility was determined as the content of water-extractable Hg, which was conducted according to Ana Teresa Reis et al.[5]. About 1 g of soil was extracted using 5 mL deionized water at 180 rpm for 24 h. The mixture was then centrifuged at 4000 rpm for 10 min. The supernatant was filtered through a 0.45-μm membrane filter. All the samples were preserved in HNO_3 (0.5%, v/v) at 4 ℃ until analysis.

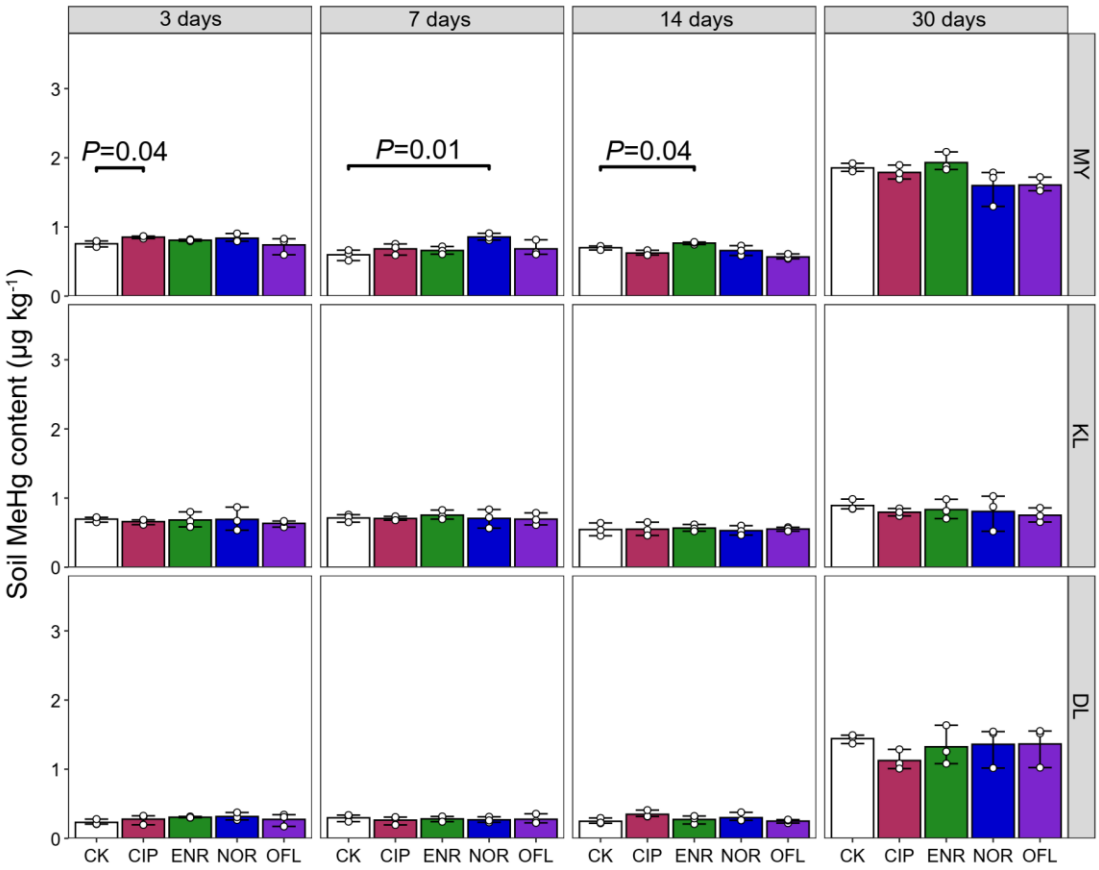
For Hg analyses, samples were oxidized with BrCl overnight at room temperature, and then Hg^{2+} was reduced using $SnCl_2$ (200 g in 100 mL 90% HCl) to purge elemental Hg. The Hg concentration was determined based on the EPA 1631 method using a cold vapor atomic fluorescence spectrophotometer (CVAFS) (Brook Rand, model III).

3. The ratios of Hg methylation and MeHg demethylation

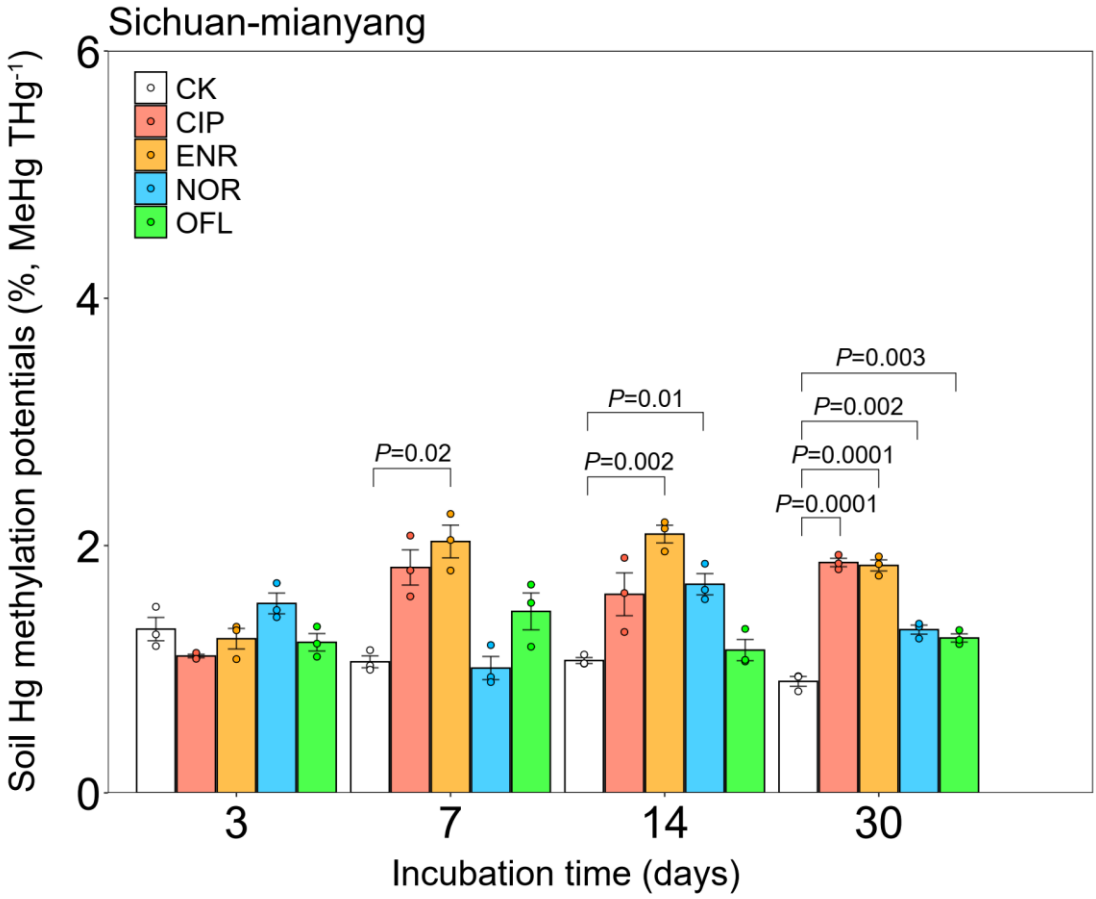
On the 21st day of incubation, all samples were transferred into an anaerobic glove box (N_2/H_2 = 95%/5%). The soil samples were spiked with $^{202}HgNO_3$ and $Me^{200}HgCl$ solution to reach a concentration of about 80% of the background contents of THg and MeHg[6, 7]. After being mixed thoroughly, all tubes were tightened with covers and incubated continuously at 25 ^{o}C for 24 h in the dark. At the end of incubation, samples

were centrifuged at 4000 rpm for 10 min and then transferred into the anaerobic glove box to separate the soil particulate and supernatant. About 0.6 g–1.0 g of the fresh soil was placed in a 15 mL centrifuge tube for MeHg extraction, as described in the main text.

The concentrations of MeHg isotopes were measured using gas chromatography coupled with inductively coupled plasma mass spectrometry (ICP-MS; NexION300, PerkinElmer) following the EPA 1630 method. The intensities of $CH_3{}^{199}Hg^+$ (internal standard, added to the soils before MeHg extraction), $CH_3{}^{200}Hg^+$ (demethylation tracer), $CH_3{}^{201}Hg^+$ (to calculate ambient MeHg), and $CH_3{}^{202}Hg^+$ (methylation tracer) according to time were recorded for the calculation of peak area and subsequently, contents of isotopic MeHg. K_m = specific methylation rate constant (in d^{-1}), and $K_m = CH_3{}^{202}Hg^+$/ (spiked $T^{202}Hg^{2+} \times t) \times 100\%$; K_d = specific demethylation rate constant (in d^{-1}), spiked $CH_3{}^{200}Hg^+/t = -K_d$ ($CH_3{}^{200}Hg^+$); t = incubation time (in days) [6].



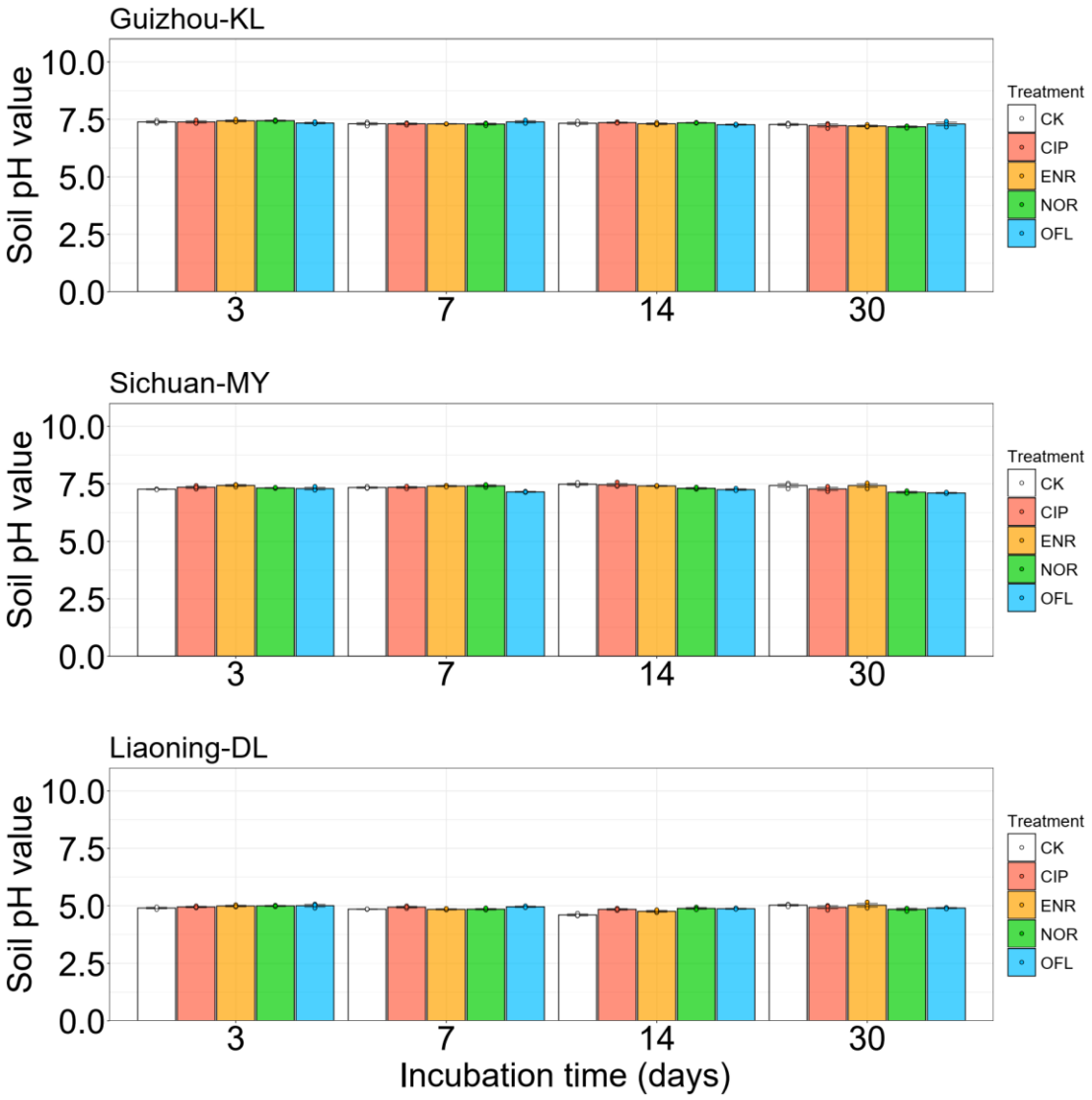
Supplementary Fig. 1 Effects of lower-dose fluoroquinolone (FQ) antibiotics (10 μg kg^{-1}) on methylmercury (MeHg) production in rice soils. KL, MY, and DL refer to paddy soils with relatively high abundance of Hg-methylating sulfate-reducing bacteria, iron-reducing bacteria, and methanogens (details in Supplementary Table 1). Treatments are denoted as CK (FQ-free), CIP (ciprofloxacin), ENR (enrofloxacin), NOR (norfloxacin), and OFL (ofloxacin), respectively. Data are presented as mean ± standard deviation (n = 3), and statistical significance was assessed using a two-tailed unpaired *t*-test.



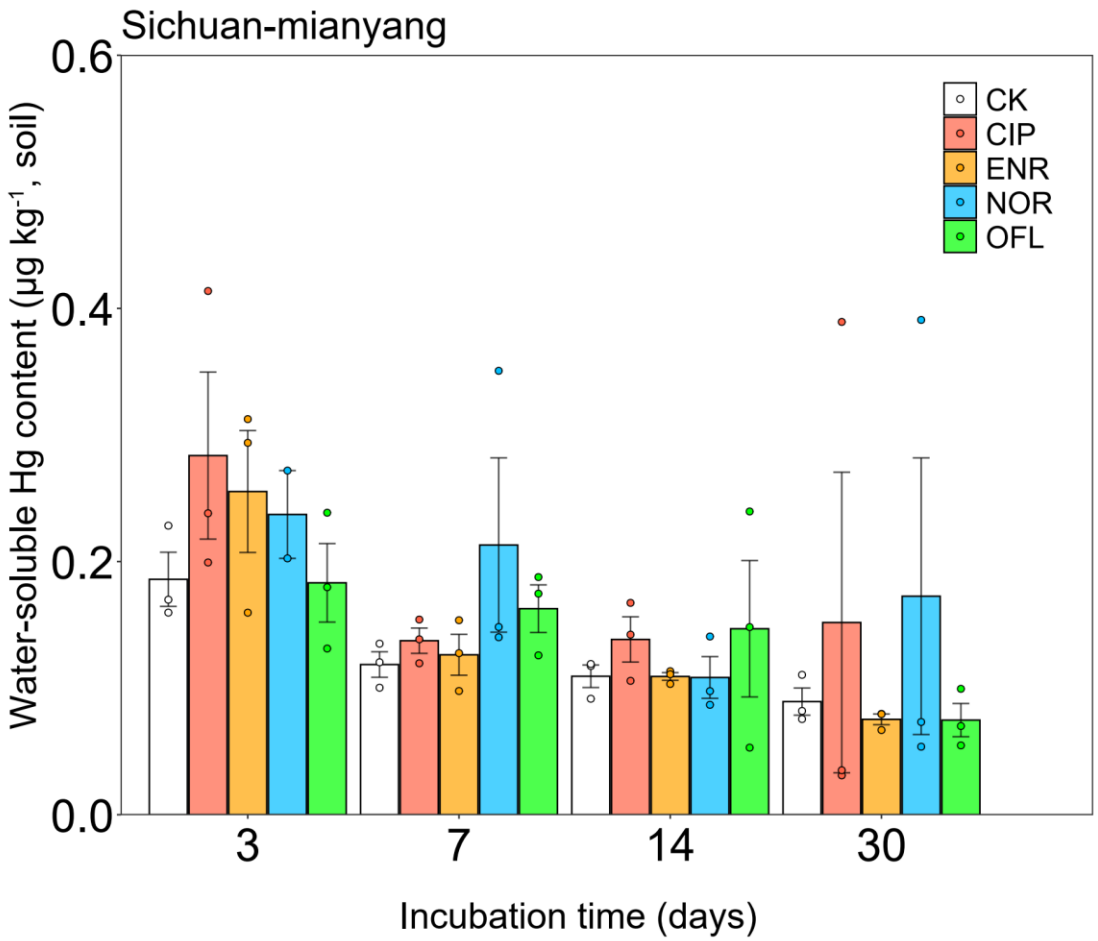
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215 **Supplementary Fig. 2 Hg methylation potentials in Mianyang soils added with or**
216 **without fluoroquinolone (FQ) addition (Batch assay I).** Treatments are denoted as
217 CK (FQs-free), CIP (ciprofloxacin), ENR (enrofloxacin), NOR (norfloxacin), and OFL
218 (ofloxacin), respectively. The different shape points represented different incubation
219 durations. Data are presented as mean ± standard deviation (n = 3), and statistical
220 significance was assessed using a two-tailed unpaired *t*-test.

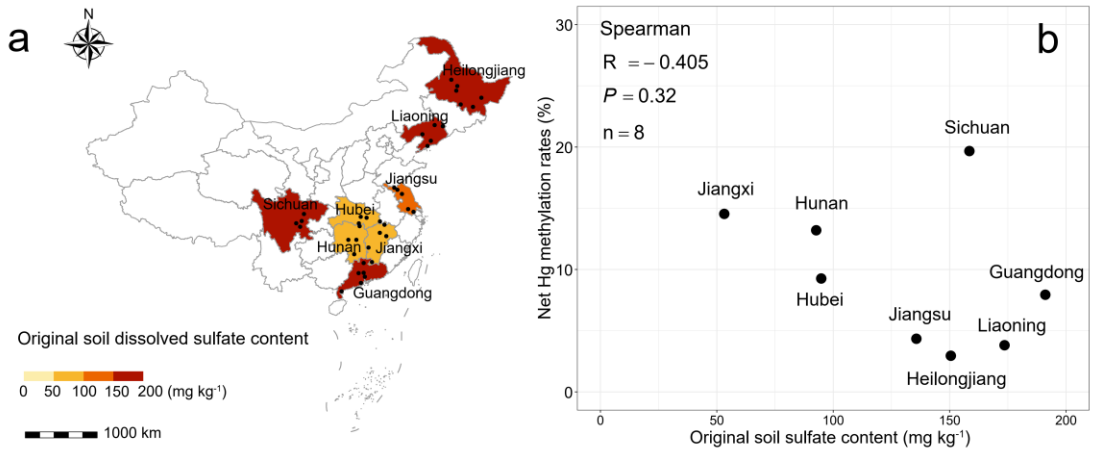
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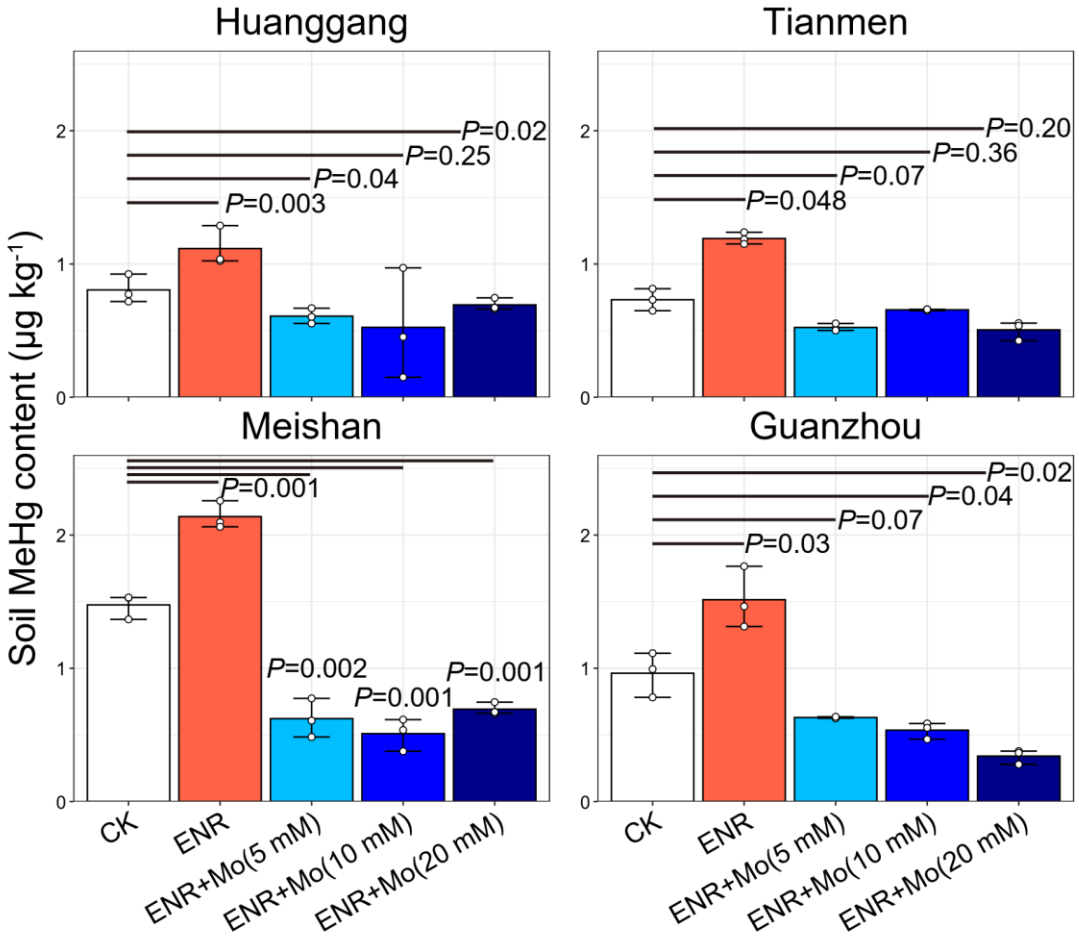
Supplementary Fig. 3 Soil pH values in the fluoroquinolone-spiked treatments (Batch assay I). KL, MY, and DL refer to paddy soils with relatively high abundance of Hg-methylating sulfate-reducing bacteria, iron-reducing bacteria, and methanogens (details in **Supplementary Table 1**). Treatments are denoted as CK (FQ-free), CIP (ciprofloxacin), ENR (enrofloxacin), NOR (norfloxacin), and OFL (ofloxacin), respectively. Data are presented as mean ± standard deviation (n = 3), and statistical significance was assessed using a two-tailed unpaired *t*-test...



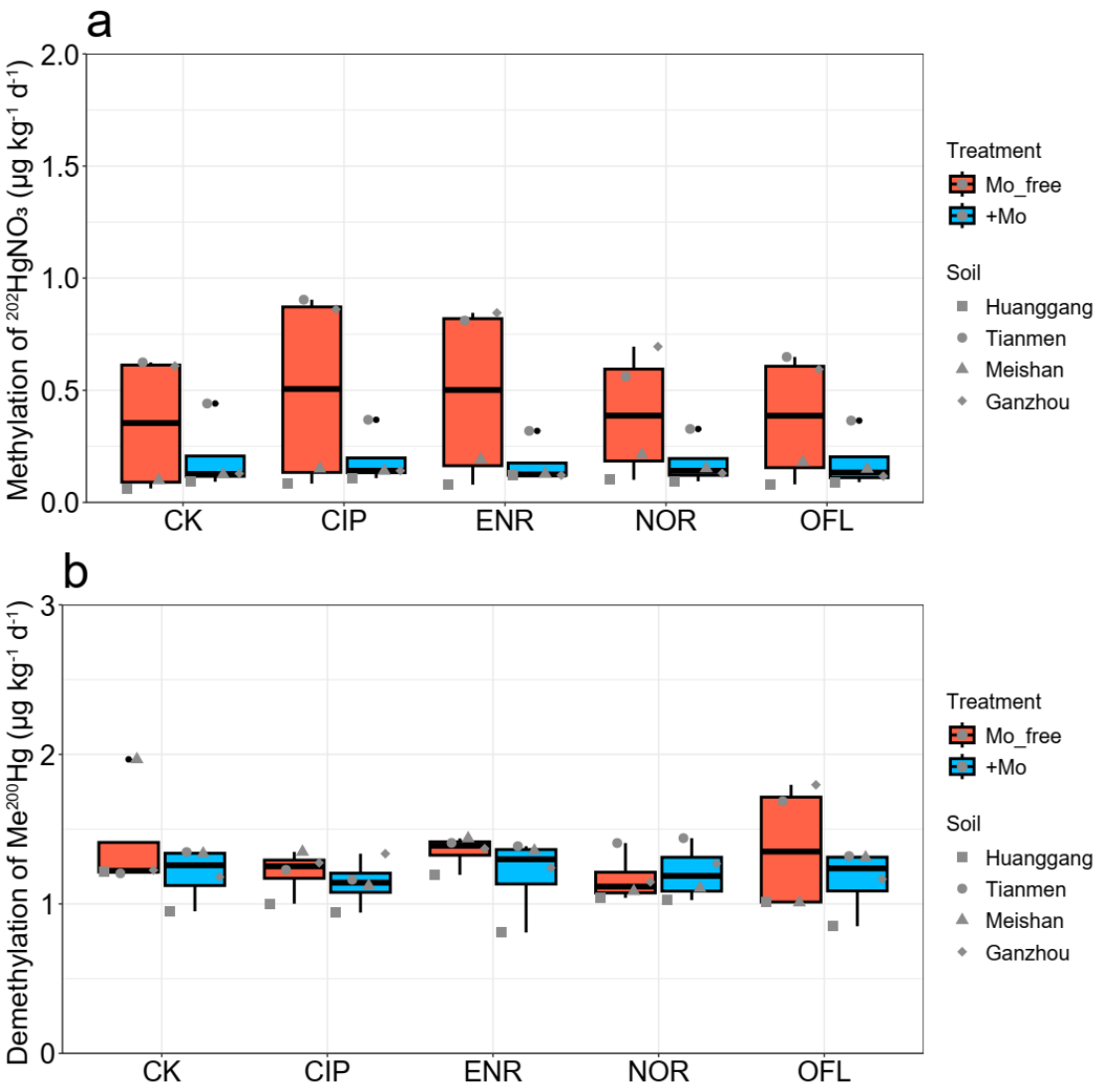
Supplementary Fig. 4 Water-extractable total Hg contents in Mianyang soils added with or without fluoroquinolone (FQ) addition (Batch assay I). Treatments are denoted as CK (FQ-free), CIP (ciprofloxacin), ENR (enrofloxacin), NOR (norfloxacin), and OFL (ofloxacin), respectively. Data are presented as mean ± SD (n = 3), and statistical significance was assessed using a two-tailed unpaired *t*-test.



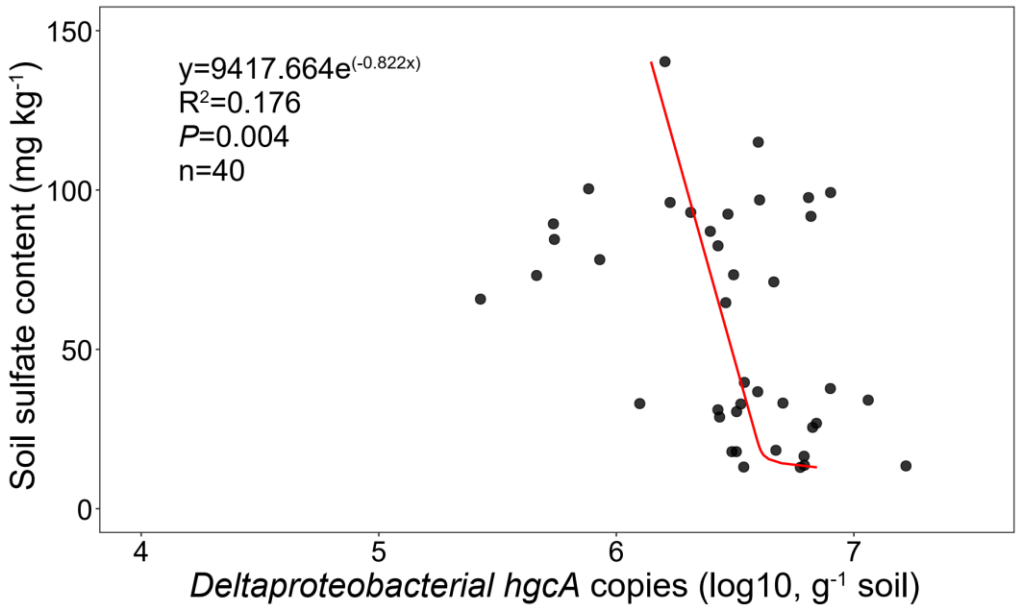
Supplementary Fig. 5 The relationship between soil sulfate content and Hg methylation potentials from eight provinces across China. (a) The original soil sulfate contents from eight provinces across China and each point represents a sampling site, the China map was generated using R (version 4.1.2). (b) The correlation between soil sulfate content and net Hg methylation rates.



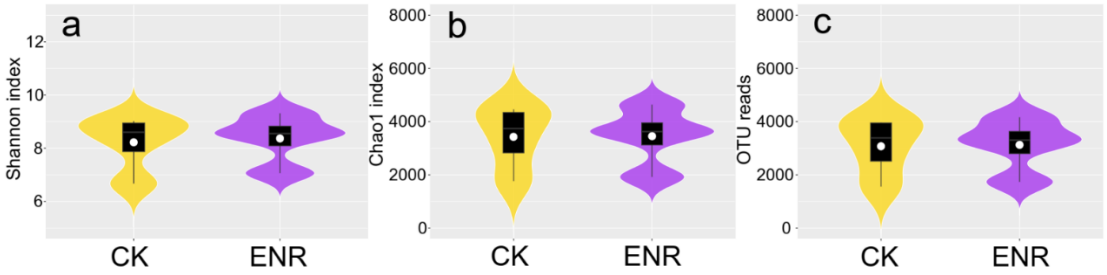
Supplementary Fig. 6 MeHg contents in enrofloxacin-treated soils spiked with different doses of molybdate (Batch assay III). Data are presented as mean ± SD (n = 3), and statistical significance was assessed using a two-tailed unpaired *t*-test.



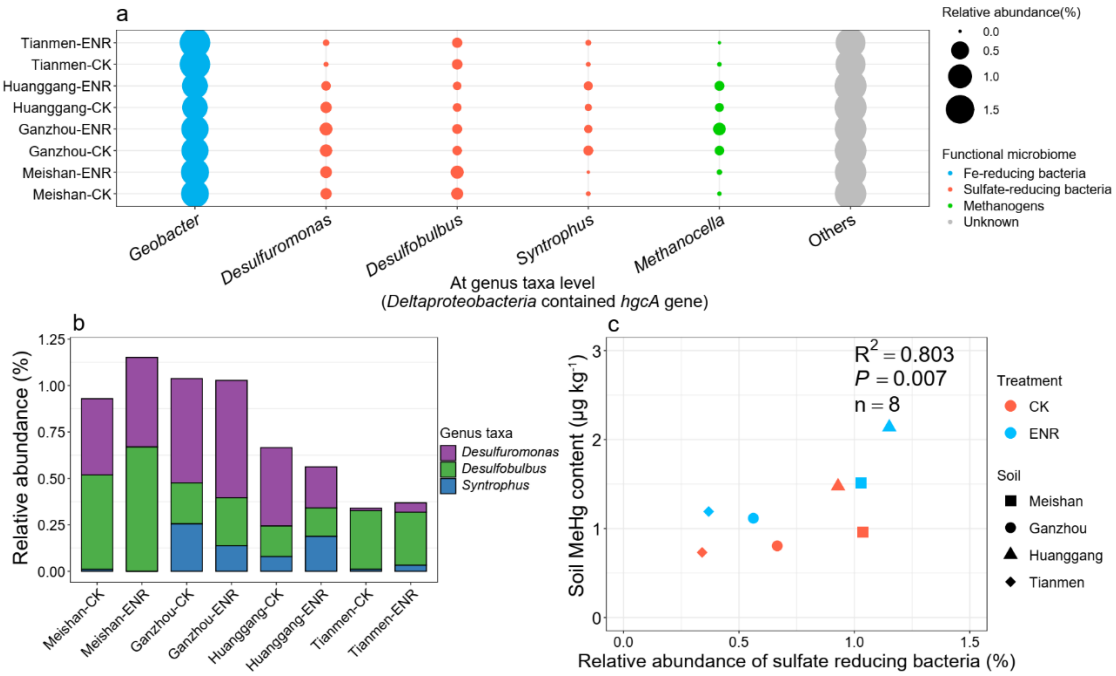
Supplementary Fig. 7 Fluoroquinolone (FQ) effects on Hg methylation and MeHg degradation spiked with isotopic Hg and MeHg in rice soils (Batch Assay III). (a), soil inorganic Hg methylation rate in the FQ-amended soils spiked with molybdate (+Mo) or in Mo-free soil controls (Mo-free); (b), soil MeHg demethylation rate in the FQ-amended soils spiked with molybdate (+Mo) or in Mo-free soil controls (Mo-free). Each bar represents the distribution of MeHg concentrations across four soils. Treatments are denoted as CK (FQ-free), CIP (ciprofloxacin), ENR (enrofloxacin), NOR (norfloxacin), and OFL (ofloxacin), respectively. In the box plot, the box shows the interquartile range (25–75%), whiskers extend to 1.5 times the IQR; the line marks the mean, and dark dots represent outliers. Statistical analysis was performed using a two-tailed unpaired *t*-test.



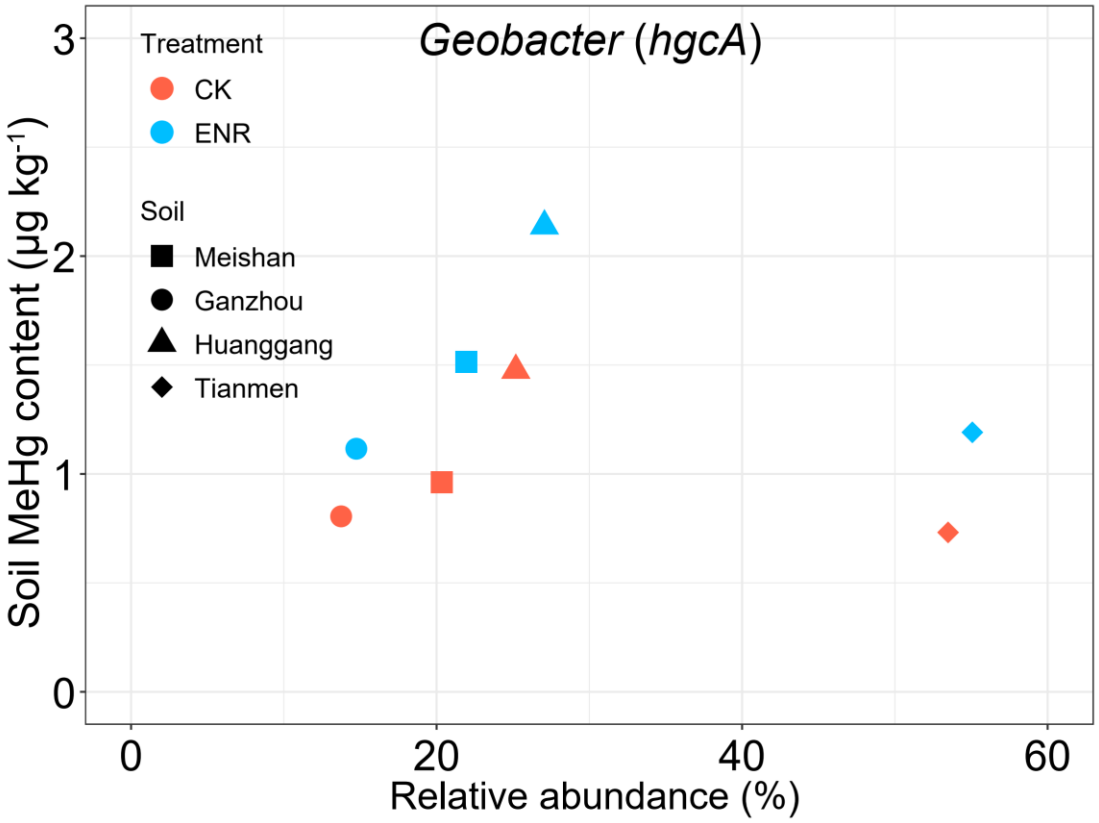
Supplementary Fig. 8 The relationship between *hgcA* gene copies of *Deltaproteobacteria* and dissolved sulfate content in soils (Batch assay III).



Supplementary Fig. 9 α-diversity indices of *hgcA*-containing *Deltaproteobacteria* in incubation soils with and without enrofloxacin (ENR) addition (based on *Deltaproteobacterial hgcA* profile; Batch Assay II). Each treatment represents the combined result values from four soils: Huanggang, Tianmen, Meishan, and Ganzhou, in the violin-box plot, the white dot represents the median; the lower and upper quartiles of the box represent the interquartile range (IQR, 25%–75%); whiskers extend to 1.5 times the IQR.

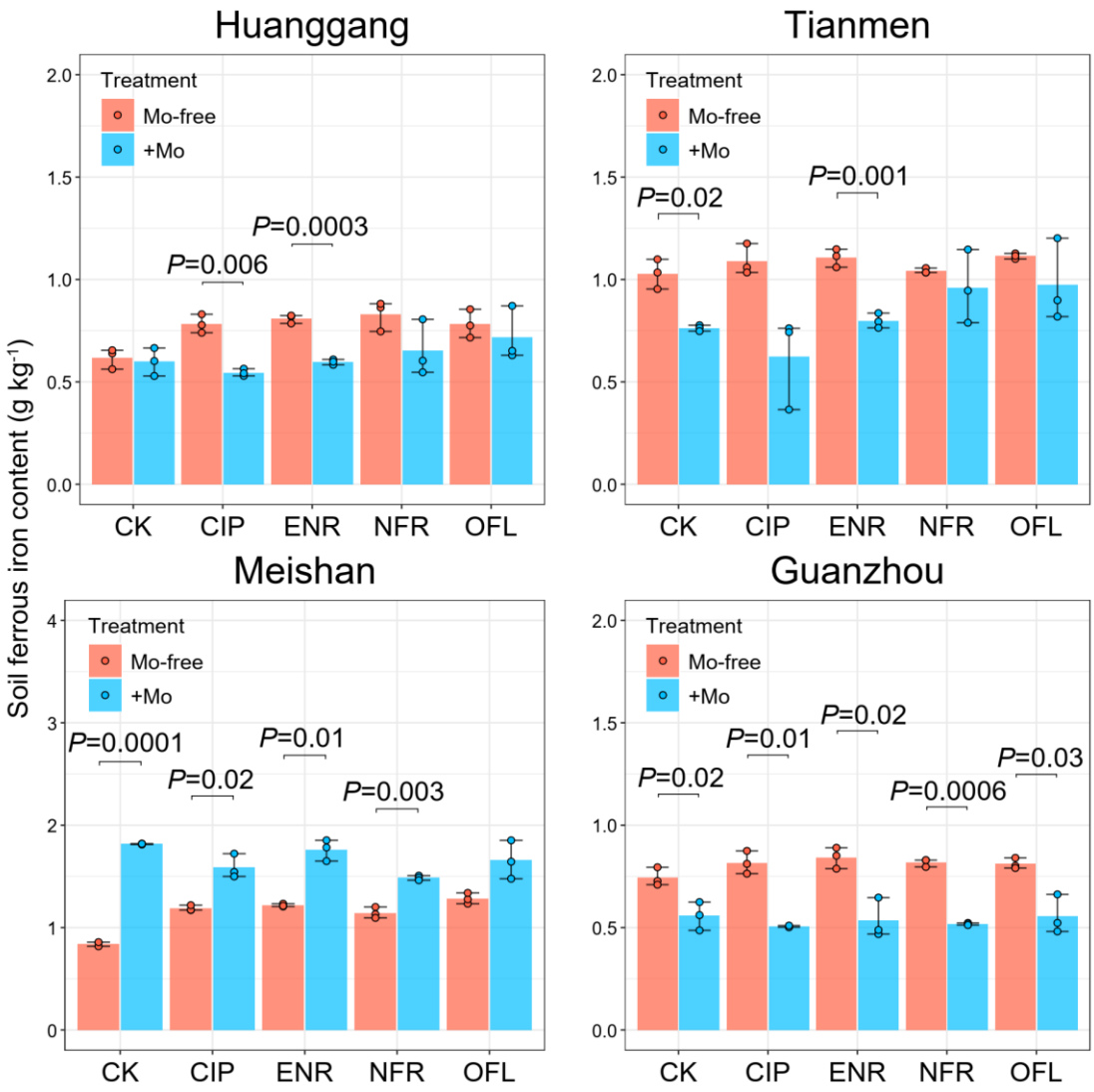


Supplementary Fig. 10 Effects of enrofloxacin (ENR) on the microbial composition of *hgcA*-containing *Deltaproteobacterial* community in paddy soils (based on *Deltaproteobacterial hgcA* profile; Batch assay II). (a) Composition of *hgcA*-containing *Deltaproteobacteria*. (b) Composition of *hgcA*-containing sulfate-reducing bacteria (SRB) across different soils. (c) Correlation between the relative abundance of *hgcA*-containing SRB and MeHg contents in soils.

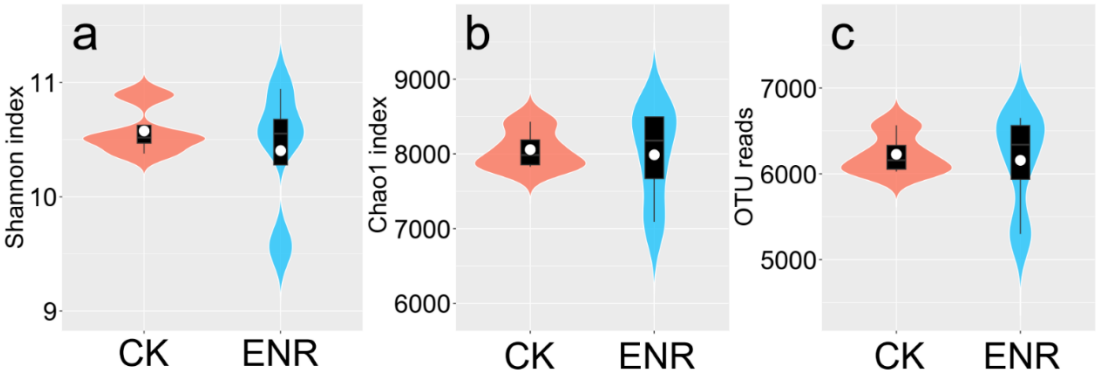


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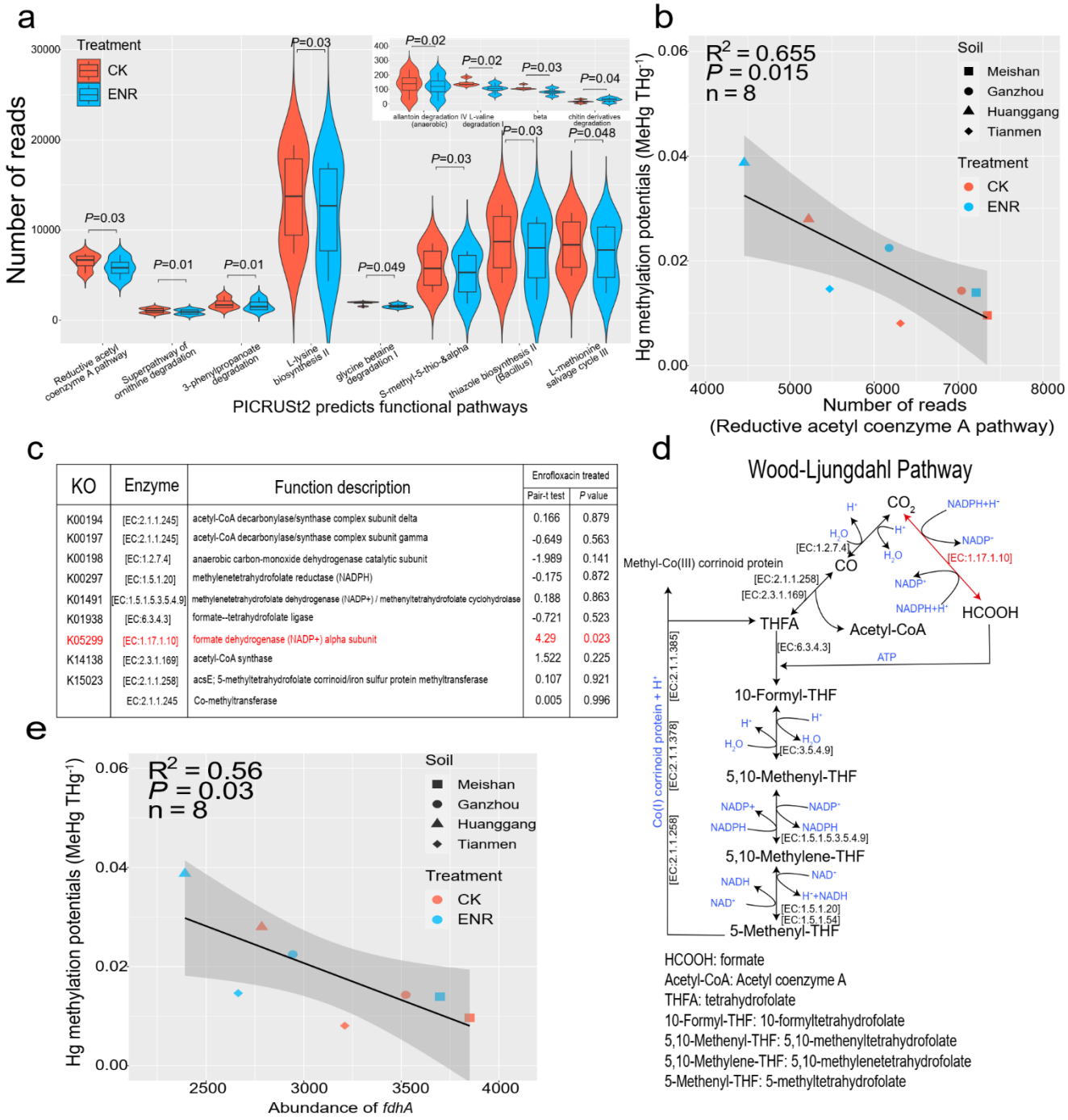
280 **Supplementary Fig. 11 Relationship between the relative abundance of *hgcA*-**
281 **containing *Geobacter* spp. and MeHg content in soils (based on**
282 ***Deltaproteobacterial hgcA* profile; Batch assay II).**



Supplementary Fig. 12 Effects of fluoroquinolones (FQs) on Ferrous contents in paddy soils (Batch assay III). The red bars represent treatments without molybdate, and the blue bars indicate treatments with molybdate addition. Treatments are denoted as CK (FQ-free), CIP (ciprofloxacin), ENR (enrofloxacin), NOR (norfloxacin), and OFL (ofloxacin), respectively. Data are presented as mean ± standard deviation (n = 3), and statistical significance was assessed using a two-tailed unpaired *t*-test.



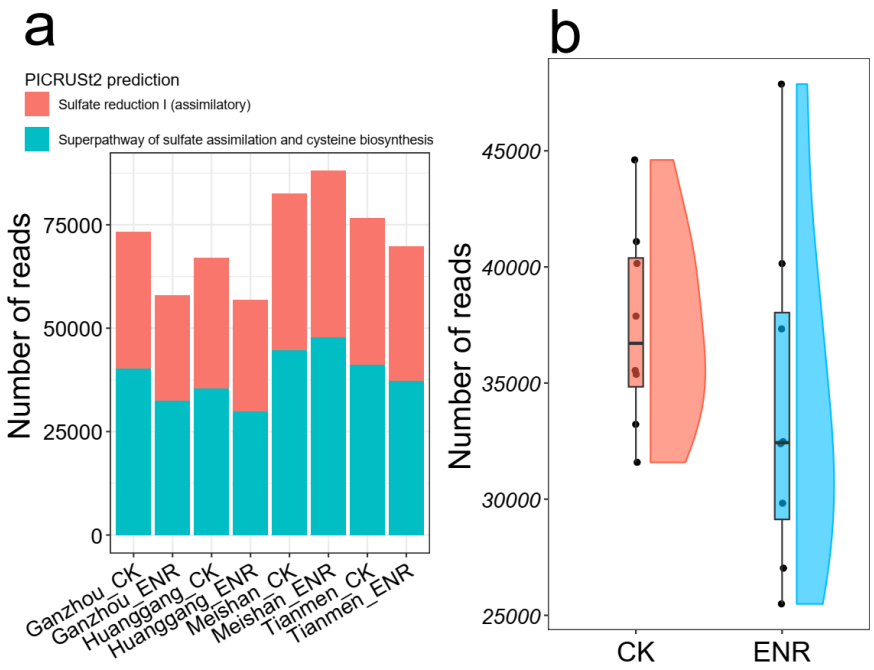
Supplementary Fig. 13 Changes in microbial alpha diversity indices and OTU read counts following enrofloxacin (ENR) addition (based on 16S rRNA genes; Batch assay II). Each treatment represents the combined result values from four soils: Huanggang, Tianmen, Meishan, and Ganzhou. In the violin-box plot, the white dot represents the median; the lower and upper quartiles of the box represent the interquartile range (IQR, 25%–75%); whiskers extend to 1.5 times the IQR.



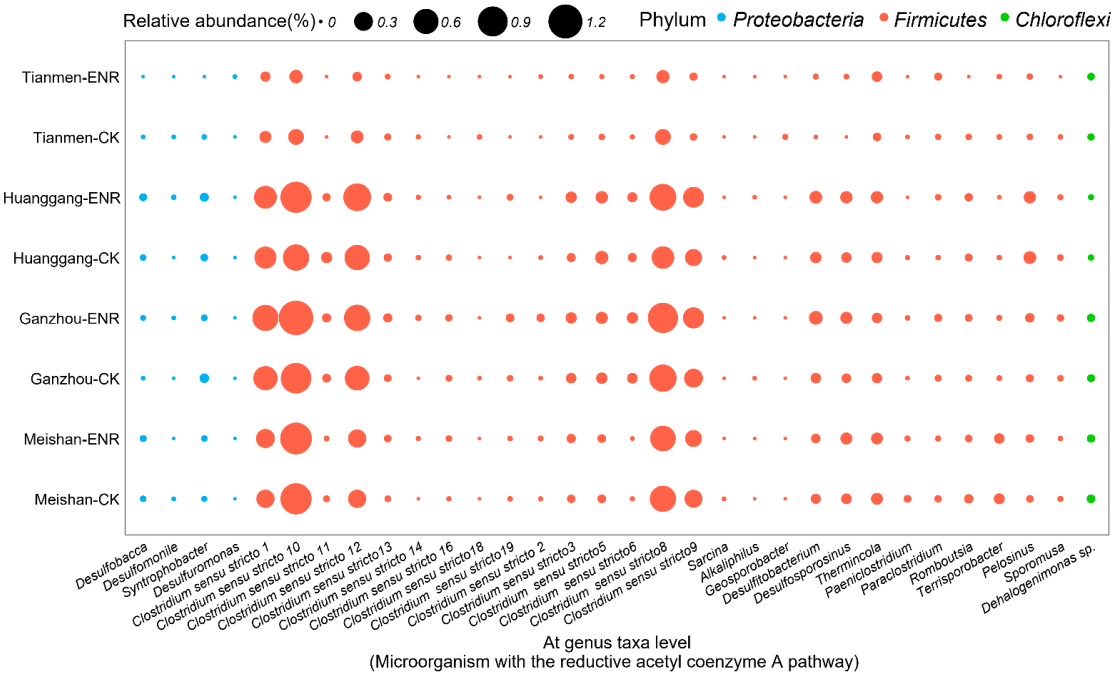
KO	Enzyme	Function description	Enrofloxacin treated	
			Pair-t test	P value
K00194	[EC:2.1.1.245]	acetyl-CoA decarbonylase/synthase complex subunit delta	0.166	0.879
K00197	[EC:2.1.1.245]	acetyl-CoA decarbonylase/synthase complex subunit gamma	-0.649	0.563
K00198	[EC:1.2.7.4]	anaerobic carbon-monoxide dehydrogenase catalytic subunit	-1.989	0.141
K00297	[EC:1.5.1.20]	methylenetetrahydrofolate reductase (NADPH)	-0.175	0.872
K01491	[EC:1.5.1.5.3.5.4.9]	methylenetetrahydrofolate dehydrogenase (NADP+) / methenyltetrahydrofolate cyclohydrolase	0.188	0.863
K01938	[EC:6.3.4.3]	formate--tetrahydrofolate ligase	-0.721	0.523
K05299	[EC:1.17.1.10]	formate dehydrogenase (NADP+) alpha subunit	4.29	0.023
K14138	[EC:2.3.1.169]	acetyl-CoA synthase	1.522	0.225
K15023	[EC:2.1.1.258]	acsE; 5-methyltetrahydrofolate corrinoid/iron sulfur protein methyltransferase	0.107	0.921
	EC:2.1.1.245	Co-methyltransferase	0.005	0.996

Supplementary Fig. 14 Reductive acetyl-coenzyme A and other related carbon metabolism pathways, and their hypothetical relationships with potential Hg methylation in the enrofloxacin (ENR) amended soils predicted by KEGG orthologs (KO) metagenome analyses through Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt2) (Batch assay II). (a) ENR effects on carbon metabolism pathways in the control and ENR-spiked soil through the KEGG orthologs (KO) predictions from PICRUSt2. In the violin-box plot,

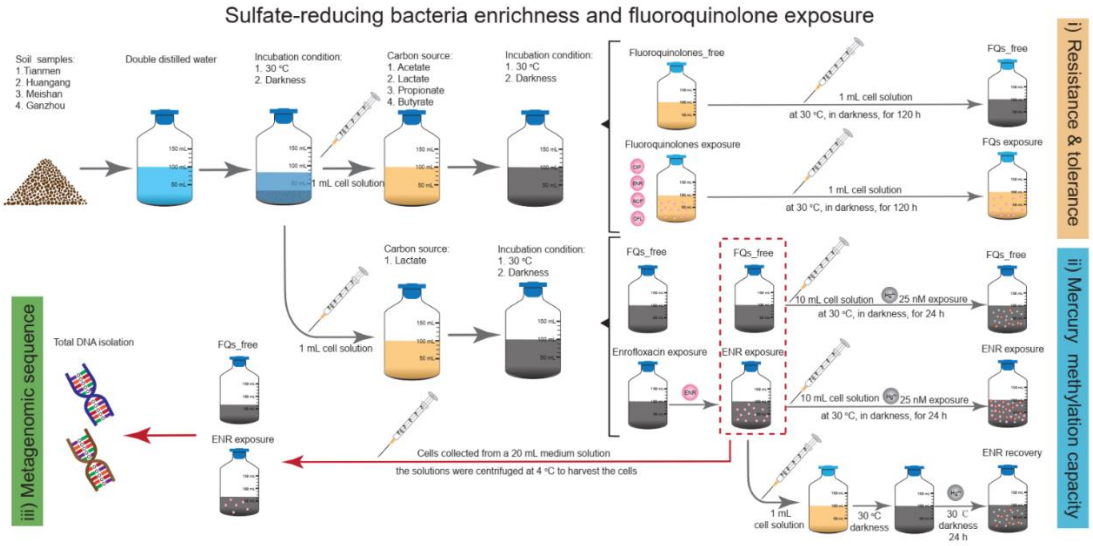
the center line represents the median; the lower and upper quartiles of the box represent the interquartile range (IQR, 25%–75%); whiskers extend to 1.5 times the IQR; vertical lines represent the standard deviation (SD) values and, and statistical significance was assessed using a two-tailed unpaired *t*-test. (b) The relationships between the abundance of the reductive acetyl-coenzyme A pathway (acetyl-CoA pathway) and Hg methylation potential (MeHg THg^{-1}, %) in different soils. (c) ENR effects on the enzymes involved in the acetyl-CoA pathway. (d) ENR effects on the one-carbon metabolism pathway (Wood-Ljungdahl pathway). (e) The relationship between the abundance of the *fdhA* gene and Hg methylation potential (MeHg THg^{-1}).



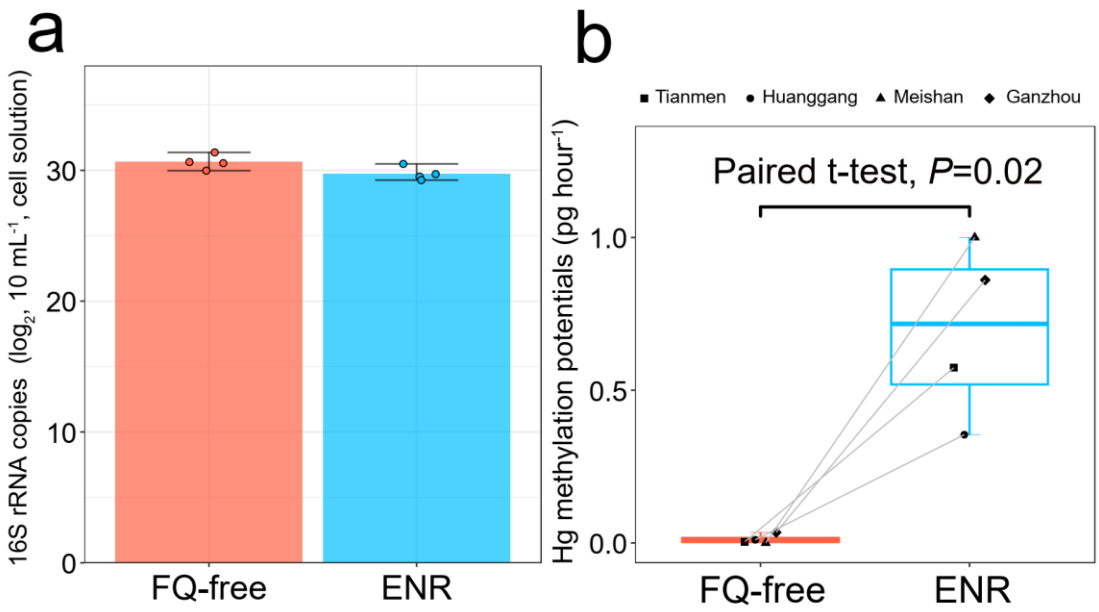
Supplementary Fig. 15 PICRUSt2 predictions of enrofloxacin (ENR) effects on sulfate-reducing functions based on 16S rRNA genes (Batch assay II). (a) Impact of ENR exposure on sulfate reduction function. (b) Change in read counts related to sulfate reduction under ENR exposure. The center line represents the median; the lower and upper quartiles of the box represent the interquartile range (IQR, 25%–75%); whiskers extend to 1.5 times the IQR; vertical lines represent the standard deviation (SD) values; the black dots represent the abundance of potential sulfate-reducing functions based on PICRUSt2 predictions for each soil sample.



Supplementary Fig. 16 Relative abundances of microbial genera associated with the reductive acetyl coenzyme A (acetyl-CoA) pathway in control (CK) and enrofloxacin (ENR) spiked soils (based on 16S rRNA genes; Batch assay II).



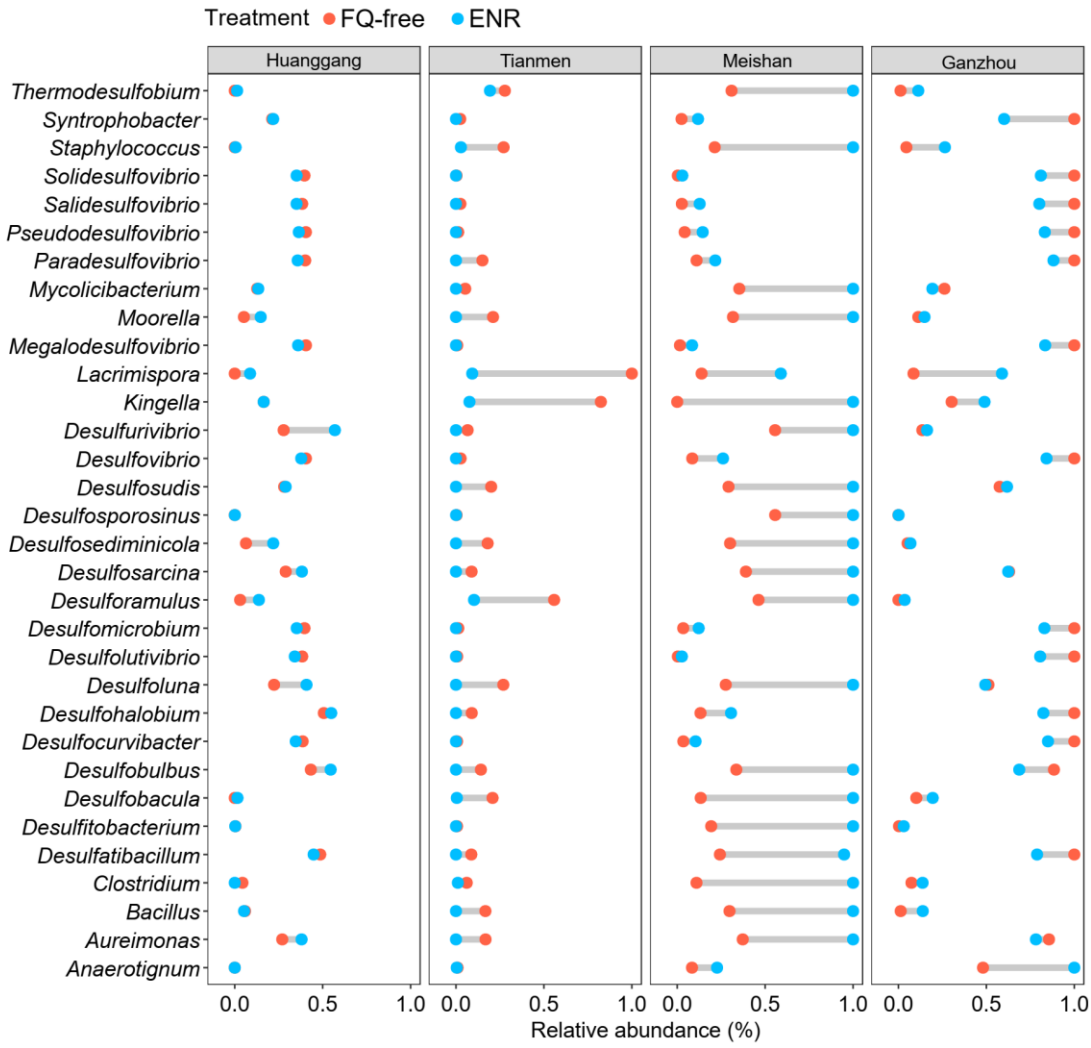
Supplementary Fig. 17 Schematic overview of the SRB enrichment from paddy soils and exposure process of enrofloxacin (ENR) (Batch assay IV). The experiment aimed to assess: (i) the tolerance or resistance of SRB to FQs, (ii) their mercury methylation capacity under ENR exposure, and (iii) functional gene expression under ENR-induced stress. SRB were enriched using SRB media supplemented with one of four carbon sources, acetate, lactate, propionate, and butyrate, and inoculated with the soils from Huanggang, Tianmen, Meishan, and Ganzhou. FQs were added at a concentration of 10 mg L^{-1}, and mercury at 25 nM. All incubations were conducted at 30 ^{o}C in the dark.



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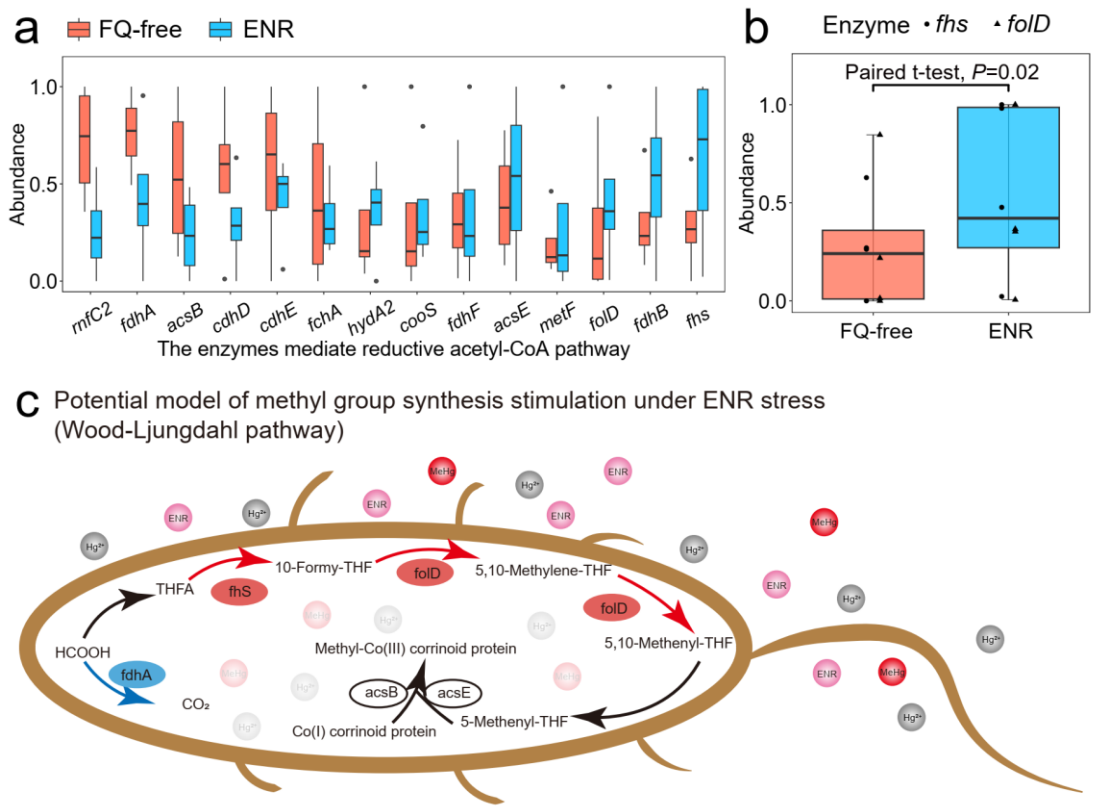
338 **Supplementary Fig. 18 Repeated Hg methylation experiment of enriched SRB**
339 **consortia under enrofloxacin (ENR) stress (Batch assay IV).** (a) The population size
340 of SRB consortia after 24 h exposure quantified by 16S rRNA gene copies, and data
341 are presented as mean ± standard deviation (n = 4). (b) MeHg synthesis after 24 h
342 incubation. In the box plot, the center line represents the median, the box shows the
343 interquartile range (25%–75%), and whiskers extend to 1.5 times the IQR. Each point
344 represents the SRB culture enriched from different soils. The SRB cultures were
345 supplemented with lactate. Statistical analysis was performed using a two-tailed paired
346 *t*-test.

347



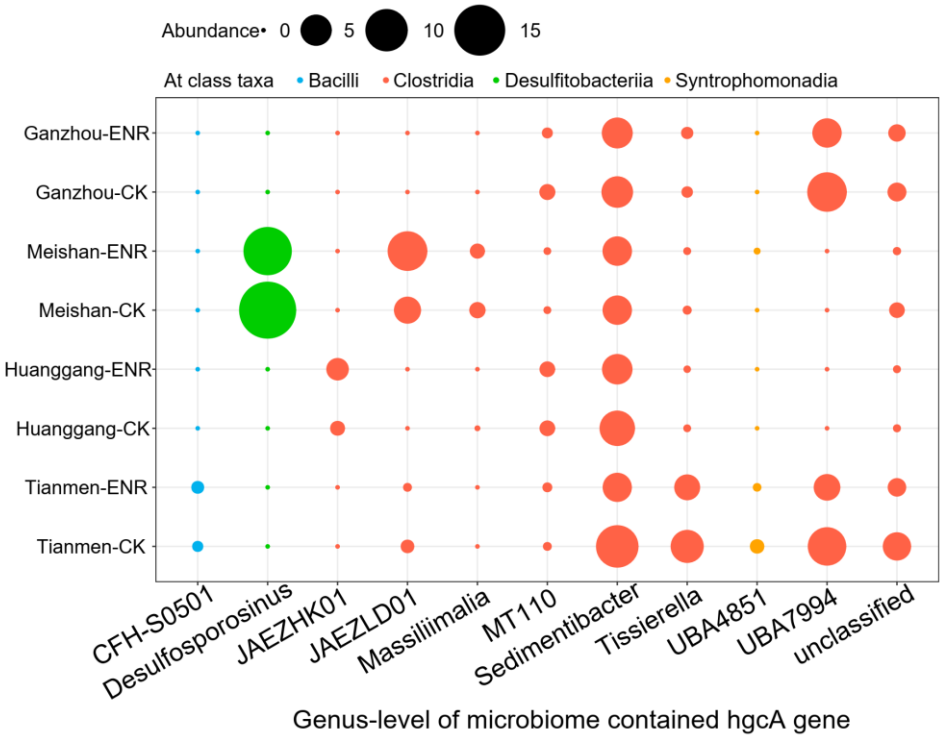
348

349 **Supplementary Fig. 19 Effects of enrofloxacin (ENR) on groups of sulfate-**
350 **reducing bacteria characterized by *dsrA*/*dsrB* genes (based on metagenomic data;**
351 **Batch assay IV).**

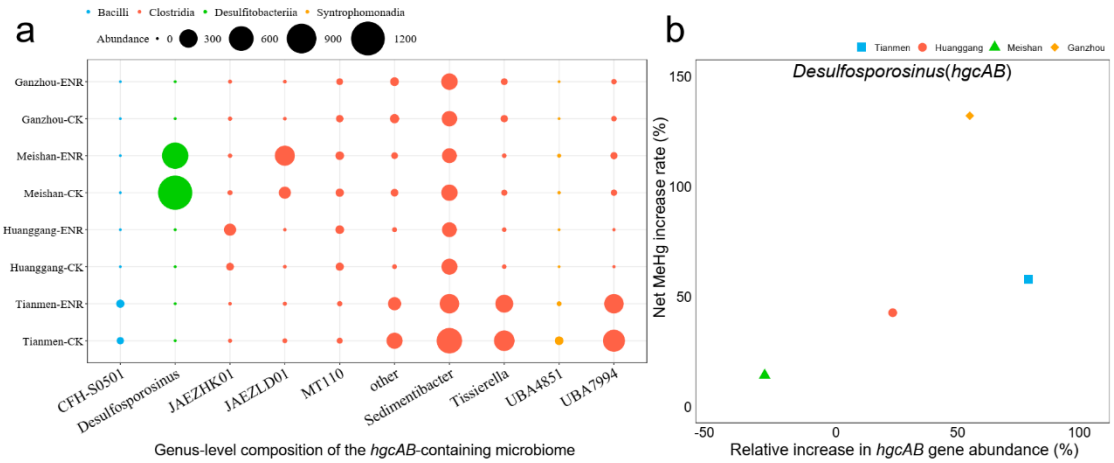


352

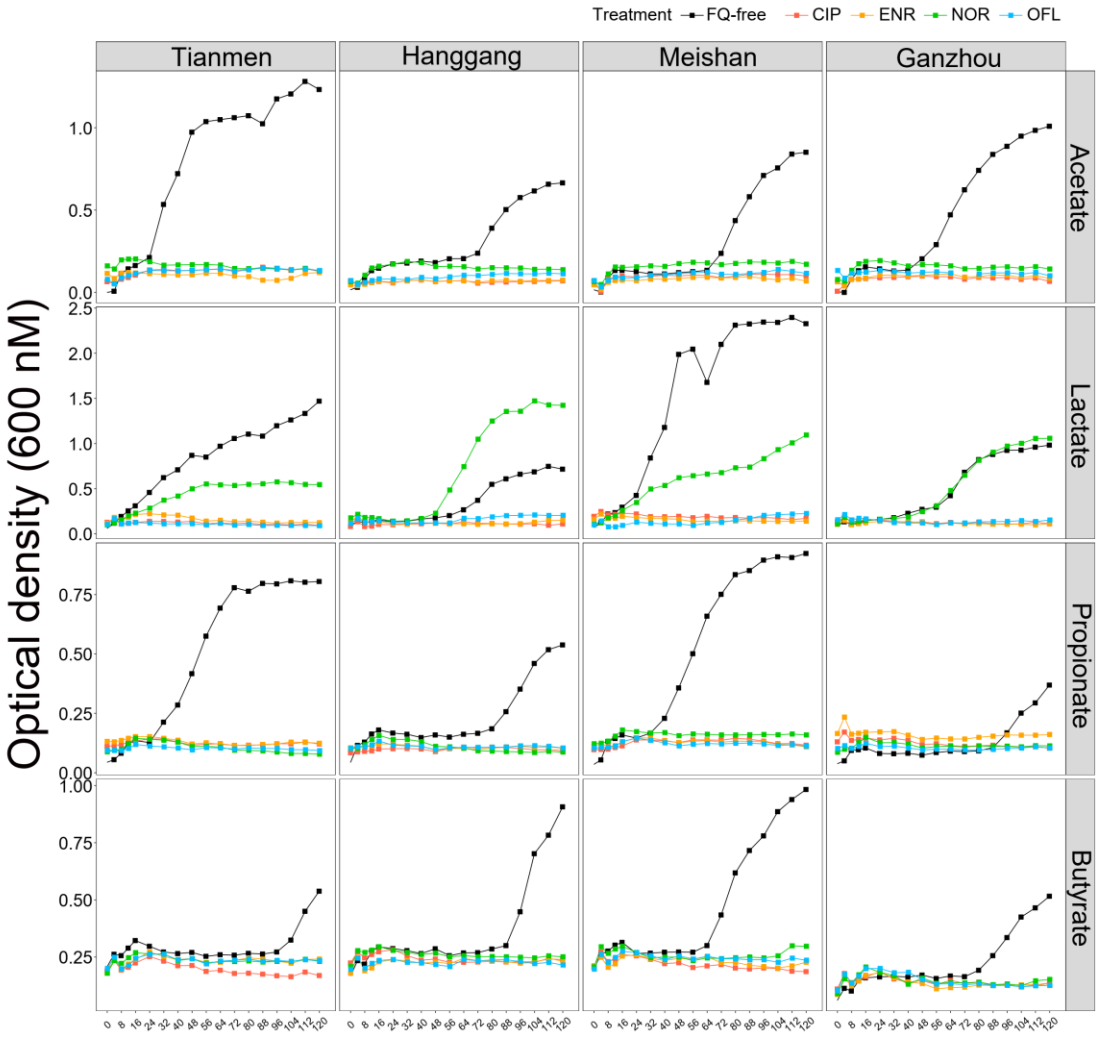
353 **Supplementary Fig. 20 Effects of enrofloxacin (ENR) stress on the reductive**
354 **acetyl-CoA pathway (based on metagenomic data; Batch assay IV).** (a) ENR effects
355 on the abundance of key genes involved in the reductive acetyl-CoA pathway, in the
356 box plot, the center line represents the median, the box shows the interquartile range
357 (25–75%), whiskers extend to 1.5 times the IQR, and black dots represent outliers. (b)
358 ENR effects on tetrahydrofolate cycling (via acetyl-CoA) in enriched SRB, in the box
359 plot, the center line represents the median, the box shows the interquartile range (25–
360 75%), whiskers extend to 1.5 times the IQR, different point symbols represent
361 individual enzyme, the same points represent the SRB culture enriched from different
362 soils, and statistical significance was assessed using a two-tailed unpaired *t*-test. (The
363 SRB cultures were supplemented with lactate.) (c) A hypothetical model illustrating
364 how ENR stimulates the reductive acetyl-CoA pathway in enriched complete SRB, with
365 red-marked enzymes indicating upregulation and blue-marked enzymes indicating
366 downregulation in response to ENR exposure.



Supplementary Fig. 21 Effects of enrofloxacin (ENR) on the *hgcA*-harboring microbial community based on metagenomics (Batch assay IV).



Supplementary Fig. 22 Effects of enrofloxacin (ENR) on the microorganisms associated with *hgcAB*-containing metagenome-assembled genomes bins (based on metagenomic data; Batch assay IV). (a) Impact of ENR on the abundance of *hgcAB*-containing microorganisms. (b) Relationship between the MeHg increase rate and changes in abundance of the *hgcAB*-containing *Desulfosporosinus*.

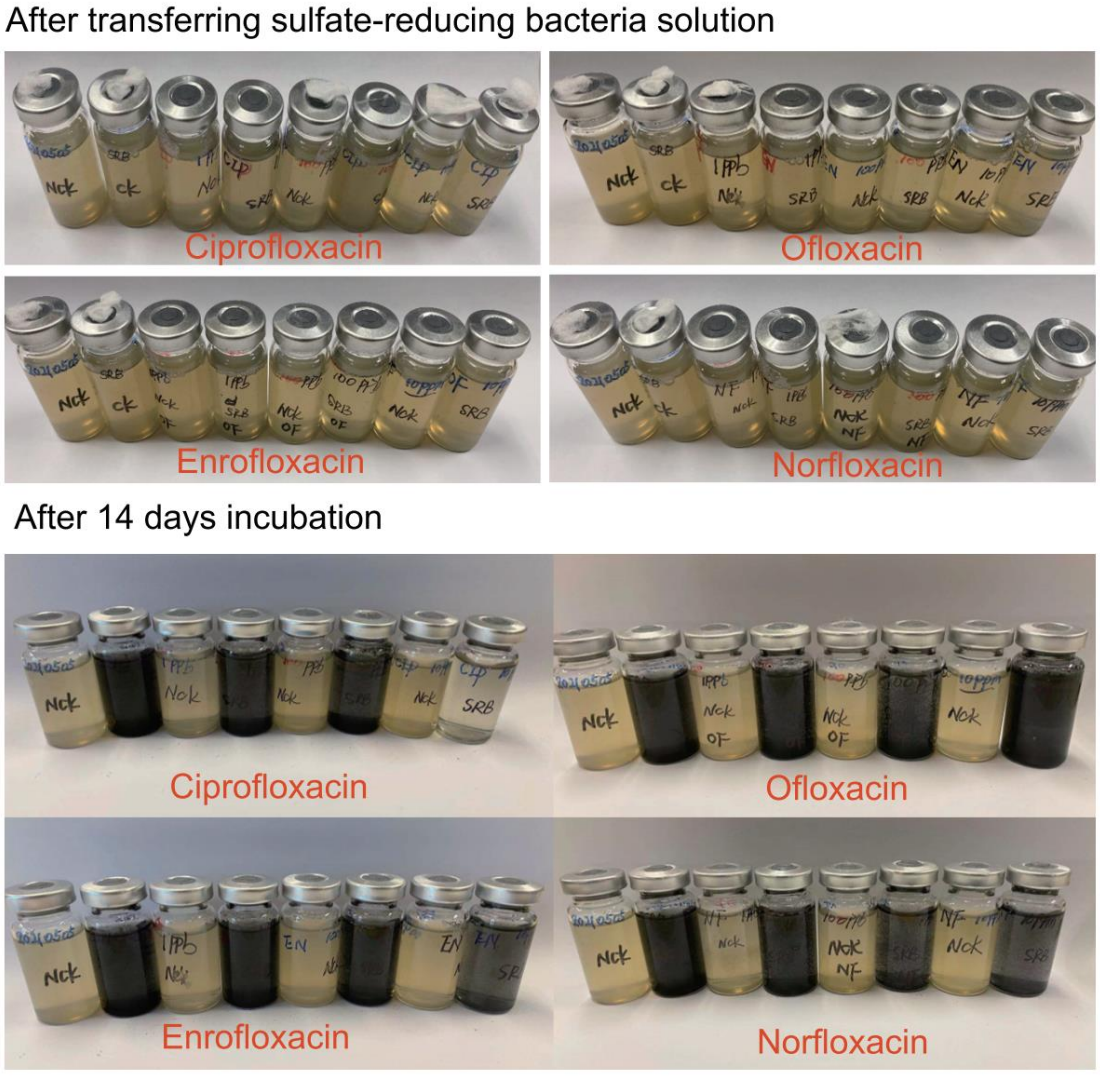


Supplementary Fig. 23 The dynamic growth of sulfate-reducing bacteria (SRB) consortia under fluoroquinolone (FQ) stresses (Batch assay IV). The panel of each row represents the SRB enriched by a medium with different carbon sources. The panel of each column represents the SRB enriched from different soils. Treatments are denoted as CK (FQ-free), CIP (ciprofloxacin), ENR (enrofloxacin), NOR (norfloxacin), and OFL (ofloxacin), respectively.

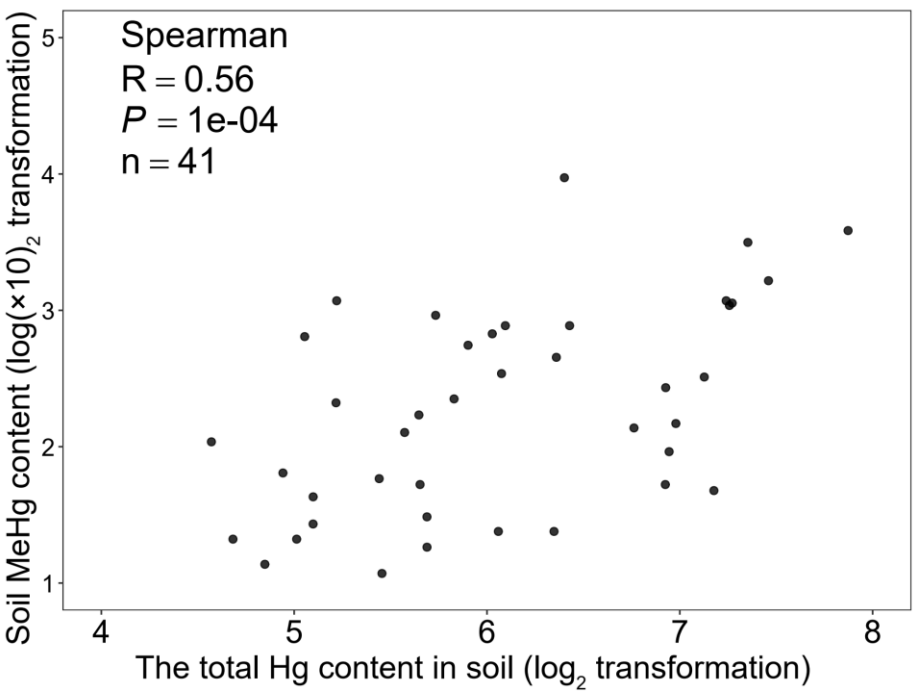
Growth statuses of sulfate-reducing bacteria while exposed to quinolone.

Treatments	1 μg/L		100 μg/L		10 mg/L	
	NCK	+SRB	NCK	+SRB	NCK	+SRB
Ciprofloxacin	-	+	-	+	-	-
Enrofloxacin	-	+	-	+	-	+
Norfloxacin	-	+	-	+	-	+
Ofloxacin	-	+	-	+	-	+

Noted: NCK represents the negative control (without SRB inoculation). "-" signals no growth of SRB, and "+" signals positive growth of SRB.



Supplementary Fig. 24 Survival of sulfate-reducing bacteria (SRB) under different fluoroquinolone (FQ) stresses in Mianyang soil (Batch assay IV). Black coloration of the culture media indicates the formation of FeS, a product of sulfate reduction.



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388 **Supplementary Fig. 25 Correlation between total Hg and MeHg contents in the**

389 **original soils across all samples.**

Supplementary Table 1 Composition of *hgcA*-containing microorganisms of three representative soils in Batch assay I*.

Soil	SRB	FeRB	Methanogens	Other
DL	4.2	74.4	1.8	19.6
KL	2.13	22.3	39.3	36.27
MY	31.5	43.3	2.51	22.69

Note: the value in the table represents relative abundance (%).

* The community results were derived from *hgcA* gene sequencing data reported in our previous study (Zhong H. et al. 2024. *Nature Food* 5: 301–311).

Supplementary Table 2 The biogeochemical properties of soil samples in the national survey.

Province	City	Longitude	Latitude	Total Hg (μg kg^{-1})	Methylmercury (μg kg^{-1})	pH
Heilongjiang	Qitaihe	130.59	45.76	33.23	0.70	6.63
Heilongjiang	Harbin	127.11	47.24	59.79	0.67	6.72
Heilongjiang	Suihua	126.97	46.66	50.31	0.33	5.81
Heilongjiang	Qiqihar	126.25	48.04	34.25	0.27	6.83
Heilongjiang	Mudanjiang	129.38	44.60	53.21	0.78	5.38
Heilongjiang	Wuchang	127.61	44.91	34.26	0.31	5.29
Liaoning	Panjin	122.07	41.12	65.23	0.71	7.43
Liaoning	Fushun	125.03	42.15	84.53	1.57	5.46
Liaoning	Tieling	123.74	42.23	37.19	0.50	7.28
Liaoning	Anshan	123.00	41.12	32.29	0.25	6.41
Liaoning	Dalian	122.81	39.64	25.68	0.25	5.28
Jiangsu	Suzhou	120.79	31.29	51.58	0.24	7.45
Jiangsu	Xuzhou	118.01	34.34	30.73	0.35	7.09
Jiangsu	Changzhou	120.01	31.63	163.4	1.13	6.25
Jiangsu	Suqian	118.46	34.08	23.76	0.41	6.48
Jiangsu	Huai'an	119.15	33.51	68.36	0.74	6.32
Hubei	Jinmen	112.20	31.04	56.85	0.51	7.25
Hubei	Wuhan	114.01	30.54	154.5	0.83	6.33
Hubei	Tianmen	113.17	30.65	81.43	0.26	7.44
Hubei	Huanggang	115.94	30.07	28.80	0.22	4.71
Hubei	Xiaogan	125.03	42.15	50.10	0.47	7.01
Hunan	Yueyang	113.01	29.46	123.17	0.39	7.37
Hunan	Yongzhou	112.22	25.91	108.6	0.44	7.24
Hunan	Loudi	111.38	27.73	176.0	0.93	5.22
Hunan	Xiangxiang	112.54	27.74	121.7	0.54	6.54
Sichuan	Mianyang	104.97	31.01	82.12	0.63	7.45
Sichuan	Zigong	104.41	29.39	121.5	0.33	7.45
Sichuan	Meishan	103.85	29.83	153.0	0.82	5.67
Sichuan	Ziyang	104.68	30.11	37.28	0.84	7.31
Jiangxi	Jiujiang	116.60	29.65	43.87	0.21	5.43
Jiangxi	Yingtan	116.86	28.20	43.44	0.34	4.93
Jiangxi	Nanchang	115.93	28.62	151.2	0.84	5.02
Jiangxi	Ganzhou	114.79	24.91	67.43	0.58	4.80
Jiangxi	Ji'an	114.29	26.75	234.3	1.20	5.02
Guangdong	Dongguan	113.76	23.03	144.7	0.32	5.39
Guangdong	Zhanjiang	110.37	21.28	66.66	0.26	5.44
Guangdong	Guangzhou	113.59	23.55	86.12	0.74	5.20
Guangdong	Foshan	112.86	23.51	51.58	0.28	5.18
Guangdong	Zhuhai	113.20	22.26	139.8	0.57	5.67
Guangdong	Shaoguan	113.59	24.79	47.60	0.43	4.77

Guizhou	Kaili	107.81	26.51	126.18	0.45	7.21

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Supplementary Table 3 Residue contents of enrofloxacin in 14 paddy soils after 21 days of incubation.

Province	Location	ENR residue contents (mg kg^{-1})	Recovery rate (%) [b]
Heilongjiang	Harbin	3.35	133
Heilongjiang	Mudanjiang	3.31	96.7
Liaoning	Tieling	3.14	64.2
Jiangsu	Changzhou	4.01	105%
Jiangsu	Suqian	4.18	118%
Jiangsu	Suzhou	3.08	90.8%
Hunan	Xiangxiang	3.99	99.3%
Hunan	Yueyang	3.18	102%
Jiangxi	Nanchang	4.53	124%
Jiangxi	Ji'an	n.d. [a]	94.0%
Jiangxi	Yingtan	4.29	125%
Sichuan	Ziyang	2.59	108%
Sichuan	Zigong	2.46	74%
Guangdong	Guangzhou	3.25	108%

[a]: n.d. represents not detectable.

[b]: The data here refers to the recovery rate of 1 mg kg^{-1} ENR during the extraction processes.

Supplementary Table 4 The net methylmercury contents (μg kg^{-1}) in soil under fluoroquinolone treatment in Batch assay II.

Province	City	CK	CIP	ENR	NOR	OFL
Heilongjiang	Qitaihe	0.77±0.03	0.78±0.05	0.74±0.06	0.82±0.09	0.82±0.04
Heilongjiang	Harbin	0.42±0.01	0.46±0.05	0.43±0.01	0.45±0.04	0.54±0.14
Heilongjiang	Suihua	0.42±0.01	0.40±0.02	0.38±0.03	0.39±0.05	0.39±0.04
Heilongjiang	Qiqihar	0.22±0.03	0.22±0.01	0.22±0.01	0.22±0.02	0.23±0.01
Heilongjiang	Mudanjiang	0.58±0.03	0.57±0.04	0.64±0.12	0.56±0.04	0.56±0.04
Heilongjiang	Wuchang	0.14±0.03	0.17±0.03	0.15±0.02	0.18±0.02	0.16±0.02
Liaoning	Panjin	0.53±0.05	0.42±0.02	0.46±0.05	0.46±0.05	0.54±0.03
Liaoning	Fushun	1.10±0.13	1.16±0.09	1.11±0.04	1.10±0.10	1.09±0.06
Liaoning	Tieling	1.29±0.05	1.50±0.26	1.22±0.06	1.36±0.06	1.65±0.51
Liaoning	Anshan	0.46±0.06	0.49±0.11	0.56±0.06	0.46±0.02	0.43±0.04
Liaoning	Dalian	0.39±0.03	0.37±0.06	0.39±0.08	0.37±0.01	0.38±0.04
Jiangsu	Suzhou	0.37±0.02	0.40±0.03	0.42±0.01	0.42±0.03	0.39±0.01
Jiangsu	Xuzhou	0.52±0.03	0.69±0.08	0.64±0.04	0.62±0.05	0.53±0.01
Jiangsu	Changzhou	2.14±0.02	2.08±0.33	1.77±0.10	1.87±0.08	2.22±0.31
Jiangsu	Suqian	1.22±0.08	1.22±0.01	1.22±0.04	1.38±0.11	1.24±0.08
Jiangsu	Huai'an	2.14±0.08	2.25±0.05	2.56±0.14	2.30±0.06	2.49±0.11
Hubei	Jinmen	0.85±0.05	0.71±0.00	0.73±0.07	0.73±0.03	1.08±0.13
Hubei	Wuhan	1.69±0.14	1.56±0.03	1.45±0.11	1.43±0.18	1.81±0.09
Hubei	Tianmen	0.66±0.02	1.29±0.04	1.19±0.04	1.26±0.06	0.91±0.08
Hubei	Huanggang	0.81±0.11	0.91±0.13	1.12±0.15	0.85±0.09	0.99±0.11
Hubei	Xiaogan	0.64±0.00	0.64±0.02	0.70±0.18	0.38±0.08	0.52±0.05
Hunan	Yueyang	1.24±0.06	1.46±0.06	1.22±0.12	1.53±0.02	1.35±0.13
Hunan	Yongzhou	0.57±0.05	0.60±0.06	0.58±0.01	0.56±0.05	0.51±0.05
Hunan	Loudi	1.04±0.04	1.19±0.14	1.32±0.05	1.37±0.13	1.26±0.07
Hunan	Xiangxiang	0.78±0.10	0.94±0.13	0.81±0.08	0.86±0.07	0.91±0.06
Sichuan	Mianyang	0.65±0.03	0.67±0.15	0.70±0.05	0.90±0.04	0.76±0.06
Sichuan	Zigong	0.79±0.03	0.87±0.07	0.95±0.07	0.78±0.03	0.83±0.04
Sichuan	Meishan	1.48±0.09	1.96±0.04	2.14±0.10	1.78±0.20	2.14±0.18
Sichuan	Ziyang	1.09±0.01	1.18±0.15	1.12±0.07	1.08±0.05	1.29±0.04
Jiangxi	Jiujiang	0.22±0.02	0.26±0.04	0.30±0.05	0.29±0.02	0.22±0.03
Jiangxi	Yingtan	1.36±0.09	1.44±0.13	1.55±0.05	1.39±0.03	1.39±0.03
Jiangxi	Nanchang	1.14±0.06	1.23±0.09	1.41±0.05	1.21±0.08	1.32±0.07
Jiangxi	Ganzhou	0.96±0.17	1.27±0.10	1.51±0.23	1.38±0.16	1.24±0.15
Jiangxi	Ji'an	3.22±0.46	3.75±0.06	3.40±0.23	3.40±0.09	3.28±0.36
Guangdong	Dongguan	0.63±0.04	0.78±0.10	0.78±0.02	0.79±0.06	0.71±0.05
Guangdong	Zhanjiang	1.18±0.03	1.49±0.06	1.39±0.05	1.28±0.04	1.31±0.08
Guangdong	Guangzhou	1.23±0.04	1.32±0.06	1.29±0.07	1.27±0.07	1.29±0.07
Guangdong	Foshan	0.28±0.02	0.41±0.05	0.33±0.03	0.36±0.05	0.34±0.06
Guangdong	Zhuhai	1.21±0.07	1.05±0.03	0.99±0.10	1.28±0.16	1.15±0.18
Guangdong	Shaoguan	0.42±0.04	0.54±0.09	0.48±0.00	0.43±0.02	0.42±0.03

406 Note: The value is expressed as mean ± standard deviation (n=3). Treatments are
407 denoted as CK (FQ-free), CIP (ciprofloxacin), ENR (enrofloxacin), NOR (norfloxacin),
408 and OFL (ofloxacin), respectively.
409

Supplementary Table 5 Increase rates of MeHg contents (%) in the fluoroquinolone-spiked soils from different sampling locations in Batch assay II.

Province	fluoroquinolone antibiotics				Provincial average
	CIP	ENR	NOR	FOL	
Heilongjiang	1.68	0.78	3.18	6.18	2.96
Liaoning	5.79	-0.56	-0.04	10.07	3.82
Jiangsu	3.68	3.31	3.01	7.37	4.34
Hubei	10.27	11.91	0.37	14.50	9.26
Hunan	15.15	7.99	19.07	10.60	13.2
Sichuan	16.81	22.81	13.72	25.34	19.7
Jiangxi	16.63	19.81	12.47	9.24	14.5
Guangdong	12.19	5.77	8.68	5.10	7.94
National average	9.9	9.3	7.2	10.6	9.25

Note: Date presented here were calculated from the mean MeHg concentrations of soils collected from each province and represent the relative changes in MeHg concentrations in FQ-treated soils compared with their respective controls. As these values were derived from averaged data, standard deviations are not available. Treatments are denoted as CIP (ciprofloxacin), ENR (enrofloxacin), NOR (norfloxacin), and OFL (ofloxacin), respectively.

Supplementary Table 6 The α diversity of *hgcA*-carrying Archaeal community (based on archaeal-*hgcA* gene profile; Batch assay II).

	Shannon	Chao_1	OTU reads
CK	8.06 ± 0.48	2319 ± 574	2217 ± 523
ENR	8.00 ± 0.45	2314 ± 266	2137 ± 293

Note: The value is expressed as mean ± standard deviation (n=4). Treatments are denoted as CK (FQ-free) and ENR (enrofloxacin), respectively.

Supplementary Table 7 The relative abundance of *hgcA*-carrying Archaea at phylum level under different treatments (based on archaeal-*hgcA* gene profile; Batch Assay II).

	At phylum taxa			
	Euryarchaeota	***Crenarchaeota***	***Thaumarchaeota***	**Others**
CK	47.24 ± 13.40	0.03 ± 0.05	0.02 ± 0.03	52.71 ± 13.3
ENR	46.53 ± 15.18	0.004 ± 0.004	0.003 ± 0.004	53.47 ± 15.18

Note: The numbers on the table represent percentages (%), and the values are expressed as means ± standard deviation (n=4). Treatments are denoted as CK (FQ-free) and ENR (enrofloxacin), respectively.

Supplementary Table 8 The relative abundance of *hgcA*-carrying Archaea at genus level under FQ treatments (based on Archaeal-*hgcA* gene profile; Batch assay II).

Genus	CK	ENR
Methanocella	13.40 ± 8.85	14.71 ± 9.77
Methanoregula	4.35 ± 5.54	2.90 ± 1.90
Methanosphaerula	0.66 ± 0.43	0.30 ± 0.26
Methanomethylovorans	0.51 ± 0.80	0.24 ± 0.30
Methanospirillum	0.05 ± 0.09	0.15 ± 0.12
Methanolobus	0.03 ± 0.02	0.08 ± 0.08
Halobacterium	0.03 ± 0.02	0.02 ± 0.02
Haloarcula	0.02 ± 0.04	0.01 ± 0.02
Thermococcus	0.02 ± 0.03	0.01 ± 0.01

Note: The numbers on the table stand for percentages (%), and the values are expressed as means ± standard deviation (n=4). Treatments are denoted as CK (FQ-free) and ENR (enrofloxacin).

435 **Supplementary Table 9 Trace elements and vitamins in the SRB enrichment**
436 **medium[8].**

	Composition
Basic medium	3.5 g L^{-1} $MgSO_4 \cdot 7H_2O$, 2.0 g L^{-1} NH_4Cl 1 g L^{-1}, 1 g L^{-1} Na_2SO_4, 1 g L^{-1} Yeast extract, 0.5 g L^{-1} K_2HPO_4, 0.1 g L^{-1} $CaCl_2 \cdot 7H_2O$, 0.5 g L^{-1}, ascorbic acid, 0.1 g L^{-1} sodium thioglycolate
Trace elements solution (1000×)	$FeSO_4$, 0.1 g L^{-1}, $CoCl_2 \cdot 6H_2O$ 0.12 g/L, $MnCl_2 \cdot 4H_2O$ 0.1 g/L, $ZnCl_2$ 0.07 g/L, H_3BO_3 0.06 g/L, $NiCl_2 \cdot 6H_2O$ 0.025 g/L, $CuCl_2 \cdot 4H_2O$ 0.015 g/L, $NaMoO_4 \cdot 2H_2O$ 0.025 g/L, $NaSeO_3$ 3.0 mg/L.
Vitamin solution (100×)	Thiamine-HCl 0.1 g L^{-1}, p-Aminobenzoic acid 50 mg L^{-1}, Vitamin B12 50 mg L^{-1}, Biotin 10 mg L^{-1}.

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Supplementary Table 10 Methylation rates (%) of $Hg(NO_3)_2$ in the enriched SRB cultures under ENR stress associated with Fig. 5g (Batch assay IV).

Treatment	SRB-enriched medium from soils			
	Tianmen	Huanggang	Meishan	Ganzhou
FQ-free	7.32	3.95	4.64	13.3
ENR	15.2	5.29	5.49	22.7
ENR-recovery	3.91	6.22	3.98	9.02

Note: Treatments included a fluoroquinolone-free control (FQ-free), ENR (enrofloxacin), and ENR removal (ENR-recovery).

Supplementary Table 11 The relative abundance of sulfate-reducing bacteria characterized by *dsrA* or *dsrB* in the enriched microbial communities (based on metagenomic data; Batch assay IV).

Treatment	Relative abundance (%)			
	Tianmen	Huanggang	Meishan	Ganzhou
CK	60.93	52.35	60.08	80.99
ENR	55.33	43.87	65.73	82.36

Note: Treatments are denoted as CK (FQ-free) and ENR (enrofloxacin).

Supplementary Table 12 The reads of fluoroquinolone (quinolone) resistance genes in the enriched SRB based on metagenomic sequencing. (based on metagenomic data; Batch assay IV).

Treatment	Soil	Quinolone-*lfrA*	Quinolone-*norB*
CK	Tianmen	n.d.	n.d.
	Huanggang	10.02	68.93
	Meishan	n.d.	15.24
	Ganzhou	n.d.	19.46
ENR	Tianmen	n.d.	n.d.
	Huanggang	2.54	23.44
	Meishan	n.d.	n.d.
	Ganzhou	n.d.	33.31

Note: The numbers in the table represent read counts per million sequences, and 'n.d.' indicates samples in which the target was not detected. Treatments are denoted as CK (FQ-free) and ENR (enrofloxacin).

Supplementary Table 13 Recovery rates of MeHg detection in water samples (Batch assay IV).

Treatment	SRB enrich from the soil	Recovery rate (%)
CK	Tianmen	103
	Huanggang	98.1
	Meishan	97.7
	Ganzhou	93.0
ENR	Tianmen	100
	Huanggang	117
	Meishan	91.3
	Ganzhou	85.3

Note: Treatments are denoted as CK (FQ-free) and ENR (enrofloxacin).

Supplementary Table 14 Total sulfur and dissolved sulfate levels in soil samples from the four locations (mM)*.

	Tianmen	Huanggang	Meishan	Ganzhou
Total S	6.25	9.38	9.38	6.25
Dissolve sulfate	0.23	0.43	0.71	0.49

* Data were reported in our previous study (Zhong H. et al. 2024. *Nature Food* 5: 301–311.).

Reference

1. Lei P, Yu RQ, Kong YQ, Bertilsson S, Tsui MTK, Jiang T, et al. Properly interpret metabolic inhibition results to identify primary mercury methylating microbes. *Crit. Rev. Env. Sci. Tec.* 2023, 53(19): 1757–1773.
2. Gilmour CC, et al. Mercury methylation by novel microorganisms from new environments. *Environ. Sci. Technol.* 2013, 47: 11810–11820.
3. Ye, M.Y., Shen, Y., West, C.C., Lyon, W.G. Analysis of ferric and ferrous ions in soil extracts by ion chromatography. *J. Liq Chromatogr R. T.* 1998, 21(4): 551–565.
4. Association AAPH. *Standard Methods for the Examination of Water and Wastewater*, 20th edn, 1998.
5. Reis, A.T., Lopes, C.B., Davidson, C.M., Duarte, A.C., Pereira, E. Extraction of mercury water-soluble fraction from soils: An optimization study. *Geoderma* 2014, 213: 255–260.
6. Hintelmann, H., Keppel-Jones, K., Evans, R.D. Constants of mercury methylation and demethylation rates in sediments and comparison of tracer and ambient mercury availability. *Environ. Toxicol. Chem.* 2000, 19(9): 2204–2211.
7. Yin, R.S., Feng, X.B., Foucher, D., Shi W.F., Zhao, Z.Q., Wang, J. High Precision Determination of Mercury Isotope Ratios Using Online Mercury Vapor Generation System Coupled with Multi-collector Inductively Coupled Plasma-Mass Spectrometry. *Chinese J. Anal. Chem.* 2010, 38(7): 929–934.
8. Atlas R.M. *Handbook of media for environmental microbiology*, 2nd edn. Taylor & Francis: Boca Raton, 2005.